

CITY SIMULATOR REBUILD

A ground-up technical design for a Cities: Skylines II-scale game

September 26, 2026

Proposed architecture | algorithms | validation | production

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Rebuilding a Modern City Simulator

A technical blueprint for a Cities: Skylines II-scale game

Prepared September 26, 2026. Design proposal, research synthesis, and implementation plan.

Preface: scope and evidence

This is how I would lead a new production with the ambition of Cities: Skylines II: an editable metropolitan region, credible transport, persistent residents and companies, a legible economy, expressive construction tools, and a city that remains pleasant to play as it grows. It is a proposed architecture, not a reconstruction of proprietary source code. It does not claim to identify the internal causes of Cities: Skylines II's bugs. Public developer descriptions establish reference features; engineering sources establish available techniques; the architecture, tuning, budgets, and production estimates below are my recommendations.

The earlier Slopmerica recordings contribute concrete usability lessons: foliage silhouettes, excessive ambient lights, tool panels that obscure construction, clipped tooltips, population-inappropriate visual activity, and poorly explained facility activation. Those are observations about Slopmerica, not evidence about Cities: Skylines II. I carry their lessons into acceptance tests for this new design.

No single document can contain finished source code, every asset, all tuned constants, every platform certification requirement, and the empirical results of a game that has not been built. This report specifies the major decisions, contracts, algorithms, implementation order, and verification work needed to turn that ambition into a production. All numerical targets are initial engineering hypotheses unless explicitly identified as arithmetic examples. They require prototype measurements on selected hardware.

The reference experience includes road tools, transit and cargo, zoning, city services, citizens, climate, and an economic simulation, as described in the publisher's [official Cities: Skylines II feature overview](#). Developer diaries describe intended or historical behavior, not a guarantee about every current build. Research was checked for this report on September 26, 2026; live documentation can change afterward.

1. Executive architecture and product contract

I would build a native, headless simulation library and connect it to an established engine for graphics, input, audio, authoring, and platform integration. The simulation would remain executable without that engine. My provisional presentation choice is Unreal Engine, using an intentionally limited rendering configuration and a custom integration layer. I would confirm or reject that choice through a measured production spike before a large content investment. Choosing an engine is consequential; its feature list alone cannot settle integration cost, CPU overhead, streaming behavior, iteration speed, and mod support.

The simulation's essential promise is that a player can explain a city's failure through a chain of observable causes. A factory cannot ship because its loading bays are occupied; the bays are occupied because the downstream rail terminal is full; the terminal is full because the receiving port has no berth capacity. The game can simplify each system while preserving that chain. It must expose capacity, demand, throughput, waiting time, and the reason an operation was refused.

Five product contracts govern the project:

- **Causality:** edits have consequences through explicit connections, resources, travel, and time. Assistance or abstraction has a declared rule.
- **Legibility:** relevant causes can be inspected from an object or a city-wide panel. Quantities carry units and time windows.
- **Scale:** large populations do not require an expensive actor, full animation, or a route search for every resident every frame.
- **Continuity:** saving, reloading, streaming, camera changes, and rendering quality changes preserve authoritative city state.
- **Depth:** transport, land use, budgets, health, logistics, and services interact through understandable constraints, with enough delays and alternatives to create planning decisions.

The initial release is an offline, single-player PC game. Sandbox, guided scenarios, difficulty settings, and a credible modding foundation belong in the initial product. Multiplayer, cinematic interiors, individual rigid-body vehicle physics, and an entire planet are later research projects. Roads, rail, utilities, housing, industry, transit, basic services, parks, terrain editing, weather, and a complete save lifecycle must function together before expanding content breadth.

1.1 Scope and performance envelopes

I would target a 32 km by 32 km region with an initially buildable subset; inactive terrain has a much smaller state footprint than developed land. Map size is an address space, not a commitment to simulate a uniformly dense metropolis everywhere. Three scenario tiers define capacity: 10,000 residents for tutorial and small-town tests; 250,000 for the primary launch metropolis gate; 1,000,000 for an experimental stress target that is not advertised until proven. Each scenario also declares roads, occupied buildings, active trips, junction complexity, route churn, cargo, and disasters. Population alone is a poor workload specification.

For the reference PC, initially assume 8 modern physical CPU cores, 32 GiB RAM, an SSD, and a discrete GPU with 12 GiB VRAM. Exact CPU/GPU models and driver versions are locked in the benchmark lab during the first milestone. Target 1440p at 60 rendered frames/s in the standard 250,000-resident scenario, 20 completed simulation ticks/s at normal speed, and visibly responsive construction. A lower tier proposes 1080p/30 rendered frames/s with reduced visual density and the same simulation rules. Hardware requirements emerge from the prototype; these are not purchase recommendations or measured capabilities.

The simulation does not silently delete trips or money to maintain those numbers. When hardware cannot sustain requested speed, the game reports achieved simulation speed, preserves tick order, and keeps camera/UI responsive. Optional user-chosen simplified simulation profiles create an explicitly different scenario ruleset and

are recorded in saves. Graphics presets never change that ruleset.

2. Engine, language, dependencies, and process boundaries

2.1 Provisional stack

Use C++20 for the portable simulation and the engine bridge, with strict ownership conventions, sanitizers, and a small allowed subset for the core. Use CMake for the headless executable and libraries, the engine's supported build system for its adapter, and a pinned toolchain per platform. Python is suitable for offline content validation, regression orchestration, statistical analysis, and report generation. Data assets compile from human-readable authoring formats into typed, versioned binary records. Runtime gameplay does not parse arbitrary text inside inner loops.

Use established implementations for compression, hashing, profiling, image decoding, and geometry predicates where their correctness, licenses, platform support, and memory behavior fit. Keep a dependency inventory and pinned checksums. Avoid inventing a renderer, compression codec, general scripting language, or physics engine. Own the city-specific contracts that make this product distinctive: networks, parcels, simulation transactions, causal explanations, and content validation.

Unreal's MassEntity documents data-only fragments, archetypes, chunk organization, processors, and deferred composition changes. Unity's Entities also documents archetype chunks and the cost of structural moves. Those are useful design references, not proof that either framework automatically makes a million-person city fast. My authoritative records live in the portable library; MassEntity or a purpose-built instance layer may represent visual proxies. I would avoid maintaining two competing authoritative ECS worlds. [Epic MassEntity overview](#); [Unity archetype concepts](#).

2.2 Module ownership

The dependency direction is deliberately one-way:

```
Authoring tools -> Content compiler -> Immutable definitions
Player UI -> Typed command queue -> Simulation kernel
Simulation kernel -> Read-only snapshots -> Engine adapter -> Display
Simulation kernel -> Checkpoints + command journal -> Storage
Headless runner -> Same kernel -> Metrics + replay comparison
```

The kernel owns identity, clocks, scheduling, deterministic random streams, resource transactions, and versioned state. Domain modules own transport, land/parcels, demographics, production/trade, utility networks, services, and environment. A query layer builds indexed read models and explanations. It cannot mutate the city. The renderer owns meshes, textures, effects, skeletal poses, and transient picking proxies. The UI owns panel state and speculative previews. Storage owns encoding, compatibility, checksums, and recovery; it does not invent missing simulation outcomes.

An early dependency test forbids graphics headers, operating-system clocks, filesystem operations, network requests, and engine object pointers from domain code. External data arrives as explicit validated inputs. Tests link the kernel into a minimal executable. That allows a five-minute city regression to run without initializing a GPU or loading every texture.

2.3 Engine decision spike

Build the same small city workload with the preferred engine adapter and a thin diagnostic renderer: terrain, 50,000 building instances, 20,000 moving visual proxies, dynamic road edits, selection, save/load, and streaming camera movement. These counts are spike inputs, not promised supported limits. Measure CPU submission, render-thread stalls, draw/dispatch counts, shader compilation stutter, memory, editor iteration, and adapter complexity. A headless baseline isolates simulation cost.

Proceed with Unreal if the adapter stays narrow, no per-resident UObject is required, stable frame pacing meets the visual target, packaging and supported mod workflows are feasible, and representative content can be authored efficiently. If it fails, examine the failing subsystem first; Unity with a native simulation plugin or another

established renderer is a fallback. A custom renderer becomes justified only by measured constraints and a funded specialist team. Freeze a supported engine version for production and qualify upgrades on a branch.

3. Time, determinism, concurrency, and authoritative state

3.1 Explicit clocks

Use a 64-bit tick number. The initial design runs 20 ticks per nominal wall-clock second at 1x. Each tick advances 0.05 transport seconds. Vehicles move in meters per transport second. A proposed day lasts 1,800 transport seconds, making each transport second equal to 48 calendar seconds. A 1 km drive at 13.9 m/s therefore takes about 72 transport seconds, displayed as roughly 58 calendar minutes. This is a deliberate compressed-calendar game model. Map size, departure windows, school attendance, cargo deadlines, and commute tolerance must be tuned together against it.

All contracts name their time domain. `TravelDuration`, `CalendarDuration`, `FiscalPeriod`, and `TickCount` are distinct types. Salaries quoted per day accrue by calendar time; route travel estimates convert from transport time; construction stages have explicit calendar durations. Calendar hours cannot accidentally be compared with transport seconds. Aging can use a separately configured demographic scale, but that conversion is saved, explained, and applied consistently to cohort hazards and education transitions. Changing a day length mid-save requires an explicit migration of schedules and rates.

At 3x, the scheduler requests 60 identical ticks per wall second. It never multiplies an integration step by three. If the machine sustains only 38 ticks/s, the UI reports 1.9x achieved speed. Pausing stops authoritative time while still allowing queries and validated edit commands at a paused transaction boundary. Rendering interpolation uses the two most recent complete snapshots and a clamped fraction; it cannot advance authority.

3.2 Numerical policy

Use signed 64-bit integer minor currency units, integer item counts, fixed-point quantities for fungible resources, and checked accumulators for balances. Use residual accumulators so repeated fractional production or interest does not disappear through truncation. Select scales after overflow analysis: millimeters for persisted network coordinates within bounded tiles, milliliters only where its precision is warranted, and coarser explicit units for huge water volumes. Integer arithmetic alone is not safe without defined overflow, rounding, and saturation rules.

Authoritative decisions use deterministic comparisons and quantized inputs. Geometry authoring can use double precision and robust predicates, then commit canonical quantized topology and coordinates. Visual shaders, interpolation, particles, and animation can use ordinary floating point. Complex scientific solvers that cannot initially guarantee cross-platform bit identity run in a controlled implementation with fixed iterations and quantized outputs; they are clearly outside the first cross-platform replay guarantee until verified.

The initial determinism promise is same executable/content/ruleset and supported CPU mode, independent of worker count and render frame rate. Cross-platform bit-identical replays are a separate gate. A patch that changes rules can migrate a save without preserving an old replay's future; replay files record the original simulation version. Never market deterministic as meaning scientifically accurate.

3.3 State layout and identities

Give every persistent entity a never-reused 64-bit logical ID within its world. Runtime handles contain a dense slot and generation for fast safe lookup. Saves store logical IDs and an explicit identity mapping, never pointer addresses or incidental memory order. A generation mismatch invalidates a stale runtime reference. Slot reuse is delayed past snapshot/read epochs; it creates a new logical ID. World and content namespaces prevent an asset ID from being confused with a resident ID.

Keep hot arrays separate from cold descriptive records. A moving vehicle's lane, offset, velocity, length, leader reference, route cursor, and state flags should be contiguous for lane processing. Owner names, history, paint scheme, and inspection text live elsewhere. A resident needs persistent household/job/school references and lifecycle state, but no transform or animation when at home. Dirty sets and indexed events trigger updates for inactive entities.

Example logical records, not ABI layouts:

```
EntityRef { slot:u32, generation:u32 }
Citizen { household, job, school, lifecycle, activity, next_event }
Trip { person_or_manifest, origin, destination, state, route, deadline }
Vehicle { trip, lane, offset_mm, speed_q, length_mm, occupancy }
Facility { definition, lifecycle, access, workforce, utility_state }
Account { owner, currency, balance_i64, last_ledger_sequence }
Command { id, requested_tick, actor, payload_version, payload }
Event { tick, system, subject, cause, outcome, related_entities }
```

Structure-of-arrays is a measured choice, not a requirement to split every field. Group fields consumed together; separate fields that inflate bandwidth without helping the loop. Avoid frequent archetype migration for ordinary state changes such as stopped/moving. Use state arrays or sparse active lists when cheaper. Count bytes touched per useful update alongside elapsed time.

3.4 Tick phases and conflict resolution

Each phase reads a documented state version and writes owned output buffers. A practical order is: validate player commands; commit network and land edits; refresh affected connectivity; process due lifecycle events; publish demand and offers; match/reserve resources; compute dispatch and route work; advance movements; resolve arrivals and transfers; apply accounting and production; update scheduled environment/service summaries; validate invariants; publish snapshot/hash.

Most systems read the prior phase's committed outputs. Cycles have explicit delay semantics: this tick's delivery can update inventory before the scheduled production phase; production changes tomorrow's order estimate rather than recursively demanding infinite work in the same tick. Financial settlement has one authoritative owner. Two systems cannot independently subtract the same stock.

```
for tick in logical_time:
    commands = canonical_order(commands_due(tick))
    apply_atomic_world_edits(commands)
    for phase in phase_graph:
        jobs = stable_partitions(phase.inputs)
        parallel_compute(jobs, private_outputs)
        proposals = canonical_merge(private_outputs)
        resolve_conflicts_and_commit(proposals)
    check_invariants_and_publish(tick)
```

Parallelism accelerates independent reads and local computations. Reductions use stable ordering or exact commutative arithmetic. A road movement partition owns its lanes; boundary handoffs become proposals resolved by deterministic junction priority. Jobs do not race to claim the last hospital bed. They submit ranked reservations to the owner, which breaks ties with declared policy and stable IDs. Work stealing can change execution order but cannot change commit order.

Route planning and large recomputations use deterministic work units such as node expansions, not 'whatever completes in two milliseconds'. Continuations and queue order are state. If an asynchronous task misses its logical commit barrier, the simulation waits or follows a predeclared logical deferral rule that is identical in replay; it does not choose different routes because a CPU happened to be faster. The renderer remains independent while a tick is late.

3.5 Randomness and delayed work

Random draws are keyed by world seed, system, entity, event kind, and a persisted draw counter. Decorative particles use a separate non-authoritative stream. A new tree effect must not shift every citizen's mortality draws. Scheduled work is bucketed by deterministic phase offsets to avoid all households shopping, aging, paying rent, or reevaluating jobs simultaneously. Correlated events such as a stadium exit remain correlated because that is part of the scenario.

An idle facility registers its next relevant deadline and subscriptions instead of polling all dependencies each tick. Dependency invalidation is explicit: road access version changed, staffing changed, inventory crossed threshold, outage began, or dispatch completed. A periodic bounded audit detects missed wakeups. Factorio's engineers describe substantial gains from stopping unnecessary updates, and also a multithreading attempt limited by memory bandwidth; the lesson is to measure work and dependencies before adding threads. [Factorio optimization case studies](#).

3.6 Multi-fidelity without camera-dependent outcomes

There are three separate decisions: what a person or business logically does, how transport resources represent that action, and how it looks on screen. The logical city never follows visibility. Inactive residents can be event-driven. Travel can use microscopic lane agents in critical networks and conserved mesoscopic queues elsewhere, selected by topology, demand, and a saved deterministic policy. Rendering may independently create only a fraction of distant bodies.

Transitions must preserve identity/manifest membership, cargo, ownership, queue order where meaningful, elapsed travel, planned arrival bounds, consumed capacity, and reservations. A region cannot release a queue merely because the camera left. The reference implementation first uses one transport representation to establish behavior; hybridization is an optimization with differential tests. If equivalence cannot be established within declared error tolerances for congestion and travel, keep the simpler model and reduce advertised scale.

Visual proxies do not create additional economic agents. A harbor's background ships have declared regional/decorative roles; domestic cargo ships carry real manifests. The renderer's density budgets control what is drawn, while gameplay totals remain inspectable. Selecting an individual may promote its visual detail, never grant it a faster commute.

4. Transport, spatial fabric, and network simulation

This chapter specifies a ground-up city builder transport system. It is a proposed architecture, not a claim about *Cities: Skylines II* internals. The authoritative simulation is a headless native C++ service. The engine receives immutable render snapshots and submits versioned edit commands; camera position, frame rate, occlusion, and rendered level of detail never select simulation fidelity or alter outcomes.

All numeric limits below are **proposed, unvalidated budgets**. They are starting hypotheses to be profiled and calibrated, not product promises.

Clock contract

The authoritative core advances 20 fixed ticks per nominal wall second at 1x. One transport tick is 0.05 transport-second; one calendar day is 1,800 transport-seconds, so one transport-second advances 48 calendar-seconds. Vehicle speed, braking, dwell, signal phases, queue age, and travel time use transport-seconds. Schedules, dispatch deadlines, business hours, and demographic events use calendar time and cross the boundary only through explicit conversion functions. Thus 1 km at 13.9 m/s takes about 72 transport-seconds, or about 58 calendar-minutes. That deliberate gameplay compression requires city scale, trip lengths, and service frequencies to be tuned together.

Wall time never changes an authoritative answer. Slow hardware delays presentation instead of changing which routes are accepted, when a signal advances, or how many agents disappear. Fast-forward changes how rapidly fixed ticks are requested, not dt .

4.1 One spatial truth, several derived views

Use a right-handed world coordinate system with horizontal integer millimetres and vertical integer millimetres relative to a map-local origin. Long calculations use signed 64-bit integers; dot products and squared distances use checked 128-bit intermediates. Presentation converts into floating-point camera-relative coordinates. This avoids long-map drift and makes save/load and replay bit stable.

The authoritative spatial database contains:

- a tiled height field plus cut/fill deltas, water occupancy, soil class, and protected-area masks;
- 3D alignment curves for roads, tracks, paths, shipping channels, and aircraft procedures;
- topology objects - nodes, corridors, junctions, platforms, berths, stands, entrances, and parcels;
- legal and physical constraints - right of way, access class, clearance envelope, grade, curvature, axle load, and construction state;
- derived acceleration structures: tile BVHs, R-trees for broad-phase queries, and compact adjacency arrays.

Robust topology matters more than sub-millimetre construction precision. Orientation, segment intersection, winding, and containment predicates must produce exact signs; generated points may be rounded to the millimetre grid with a deterministic tie rule. CGAL documents this useful division between exact predicates and inexact constructions, and also warns that exactness costs performance ([CGAL exact computation guidance](#), [CGAL kernel predicates and constructions](#)). We would implement a small audited predicate kernel or use a compatible library in offline/build code.

Terrain and earthworks

Each terrain tile stores base elevation, user deltas, material, groundwater, and a monotonically increasing `terrain_version`. A corridor alignment samples neither the rendered mesh nor physics colliders. Its design profile is authoritative; the terrain mesh is derived.

For every station s along an alignment, calculate design elevation $z_d(s)$, existing elevation $z_t(s)$, and earthwork depth $d(s) = z_d(s) - z_t(s)$. Integrate deterministic cross sections to estimate cut and fill:

$$V \sim \sum_i [((A_i + A_{i+1})) / 2] * (s_{i+1} - s_i)$$

$$A_i = f(d_i, \text{formation_width}, \text{side_slope})$$

The editor solves a constrained vertical profile: endpoints and player control points are hard constraints; maximum grade, minimum vertical-curve length, clearance, drainage, and tie-in smoothness are inequalities. Straight grades connect through parabolic vertical curves, a standard road-design pattern documented by FHWA ([FHWA horizontal and vertical alignment summary](#)). If the solver cannot satisfy constraints, it returns named violations and the smallest conflicting station interval rather than silently deforming the asset.

Bridge and tunnel classification is geometric, not cosmetic. Consecutive stations become AT_GRADE, EMBANKMENT, CUT, BRIDGE, or TUNNEL spans according to earthwork depth, clearance, water, geology, and player intent. Bridges generate deck, piers, abutments, navigation clearances, and structural cost intervals. Tunnels generate portals, bore envelopes, ventilation/emergency metadata, spoil volume, and subsurface collision checks. Crossing corridors connect only through an explicit junction or transfer object; grade-separated envelopes may intersect in XY while remaining disconnected.

Horizontal alignment and corridor cross section

A road reference line is a sequence of line, circular arc, and clothoid-like transition primitives with continuous position and tangent, plus a vertical profile and superelevation profile. This resembles the separation of reference geometry, elevation, lanes, junctions, objects, and signals in the public ASAM OpenDRIVE model; OpenDRIVE also requires consistent predecessor/successor relationships and uses junctions where linkage is ambiguous ([ASAM OpenDRIVE 1.8.1 specification, road-linkage rules](#)). It is a reference model, not a mandated file format.

Cross sections are ordered bands: verge, sidewalk, cycle track, parking, vehicle lanes, median, rail, drainage, and shoulders. A band owns a piecewise width polynomial, direction, access set, speed rule, surface, and lateral offset. Transitions such as lane drops are explicit taper spans. The compiler samples only for mesh output; topology and motion evaluate the analytic alignment.

Blocks and parcels

Road rights of way are buffered into watertight polygons and unioned with water, rail, and protected masks. The residual planar faces are blocks. Parcel generation operates per block with a deterministic seed derived from stable block ID:

- 1 simplify only within a topology-preserving tolerance;
- 2 identify buildable frontage segments and access points;
- 3 recursively split by target frontage/depth, respecting minimum area and aspect limits;
- 4 reserve rear/service easements and pedestrian passages;
- 5 merge landlocked or unusable fragments into the best adjacent parcel by shared-edge score.

A parcel has stable ID, polygon, frontage intervals, access connectors, buildable envelope, zoning state, and lineage. A road edit may split or merge parcels; lineage maps old IDs to new IDs so ownership, value history, and pending construction can migrate deterministically. Every occupied parcel must have at least one valid access connector or an explicit stranded status that drives simulation consequences.

4.2 Versioned editing and network compilation

The editable network and the runtime network are different representations. The editor retains rich curves and design intent. The runtime compiler emits tightly packed structures of arrays (SoA) for movement.

```
struct NetworkTxn {
    TxnId id; PlayerId author;
```

```

    NetworkVersion expected_base;
    vector<EditOp> ops;           // add/split/move/delete/change-rule
};

struct LaneSoA {
    vector<LaneId> id;
    vector<NodeId> from, to;
    vector<uint32_t> length_mm, speed_mmps;
    vector<uint64_t> access_mask;
    vector<uint32_t> geom_begin, geom_count;
    vector<uint32_t> successor_begin, successor_count;
    vector<uint32_t> occupancy_head;
};

struct MovementSoA {
    vector<MovementId> id;
    vector<LaneId> in_lane, out_lane;
    vector<uint64_t> conflict_bits_begin;
    vector<ControllerId> controller;
    vector<uint32_t> storage_mm, saturation_milli_vph;
};

```

Apply a transaction to a private copy-on-write region, validate geometry and topology, compile affected tiles, then publish atomically at a tick boundary. The response is accepted(new_version, remap) or rejected(errors). Never expose a half-built junction. If expected_base is stale, replay commutative changes where provably safe; otherwise return a conflict containing changed stable IDs.

Compilation stages are deterministic and ordered by stable ID: snap legal endpoints; split curve intersections; classify grade separations; generate corridor bands; synthesize junction surface; create lane endpoints; enumerate legal movements; build conflict zones; create pedestrian crossings and stop lines; validate; then update routing partitions. The junction generator minimizes curvature, crossing length, and overlap while satisfying turn radius, lane continuity, and crosswalk setbacks. It must permit manual overrides because automatic lane correspondence has no single correct answer.

Key invariants after every publish:

- every live lane endpoint has a valid predecessor/successor or an intentional terminal reason;
- all successor relationships are reciprocal;
- each movement is physically continuous, mode legal, and attached to a conflict-control policy;
- no two stable live objects share a logical ID, and a tombstoned logical ID is never reused within a save; compact physical slots may be recycled only through generation-checked handles that cannot alias the old logical object;
- disconnected geometry is never inferred as connected merely from proximity;
- a demolished edge cannot retain occupancy after its closure protocol completes;
- parcel access, stop access, depot access, and emergency access resolve to live connectors.

4.3 A common multimodal graph

The canonical graph is multilayered. Walking connectors join parcel doors, sidewalks, crossings, station concourses, platforms, parking aisles, and building interiors. Cycling, road lanes, rail blocks, waterways, taxiways, and air procedures are separate layers. Transfer edges express parking, boarding, alighting, loading, security, elevators, and mode changes. They carry capacity, eligibility, time, money, discomfort, and reliability costs.

An agent minimizes generalized cost, not distance:

$$C(P) = \sum_{\text{edges } e \text{ in } P} [t_e + \alpha_k \cdot \text{fare}_e + \beta_k \cdot \text{walk}_e + \gamma_k \cdot \text{transfer}_e + \rho_k \cdot \text{risk}_e] + \epsilon(\text{agent}, P)$$

Coefficients vary by population or shipment class. `epsilon` is stable per agent and alternative so rerouting does not make preferences flicker. Edge time comes from a lagged, smoothed observation plus known incidents and schedules; agents cannot see the future. Replanning has hysteresis: switch only if improvement exceeds a threshold plus sunk deviation cost.

Hierarchical routing

Partition each mode graph into cells with boundary portals. Inside a changed cell, run bounded A* or Dijkstra. Above it, query a customizable hierarchy with shortcuts storing witness paths. Contraction Hierarchies demonstrate that preprocessing road hierarchy can reduce exact queries to a small search while retaining path unpacking (Geisberger et al., *Exact Routing in Large Road Networks Using Contraction Hierarchies*). Topology edits rebuild affected cells and ancestor separators; weight updates recustomize shortcut costs without rebuilding unchanged topology.

Transit uses timetable rounds over routes, then combines access and egress paths. RAPTOR scans each route at most once per transfer round, supports Pareto criteria such as arrival time and transfers, and does not depend on heavy preprocessing, which suits changing schedules (Delling, Pajor, and Werneck, *Round-Based Public Transit Routing*). Our internal schedule schema should export and import GTFS-like stops, routes, trips, stop times, calendars, and transfers; the canonical GTFS overview confirms those are the core public schedule concepts ([official GTFS overview](#)).

Route-cache keys include origin cell, destination cell, mode/access profile, departure-time bucket, topology epoch vector, cost-field epoch, and schedule epoch. A cache entry remains valid only when every traversed partition epoch matches. A changed edge invalidates reverse indices for paths using that partition; it does not flush the world. Before executing the next edge, an agent verifies the edge and its access rules. Failure triggers a safe-stop state and a bounded repair query.

Budgets and fair backpressure

Pathfinding is work-conserving but bounded by deterministic work units. A **proposed, unvalidated budget** is 32,768 edge relaxations and 512 hierarchy-unpacking operations per authoritative tick, with at most four ticks of debt and one-quarter of a tick's quota repaid per later tick. Requests enter deficit-round-robin queues by class: emergency, already-travelling, freight/transit operations, service dispatch, and discretionary trip creation. Each queue receives a minimum quantum; unused quota is borrowed. Within a class, age raises priority. Wall-clock milliseconds are telemetry only.

Jobs may calculate on background threads, but results publish only at a named logical barrier. The tick schedules an exact, deterministic set of work units from the quota. Every scheduled unit must finish; if a worker is late, the logical barrier blocks rather than changing simulation state. Work beyond the quota is never launched for that tick and remains in a persisted continuation queue with its original sequence number, frontier, and visited state. Thus hardware speed changes wall-clock progress only, never which query completes on a simulation tick.

When demand exceeds budget, new departures wait in an explicit `PlanningQueue`; people and cargo remain at their origin and their waiting time accumulates. Existing travellers retain their current valid route or stop safely. No agent teleports, despawns, or receives a straight-line substitute. Metrics expose queue length, oldest age, searches expanded, cache hits, repair rate, and budget debt. A **proposed, unvalidated gate** is p99 ordinary planning wait below 10 simulated seconds in the reference metropolis and zero starvation over a 24-hour run.

4.4 Microscopic movement and regional flow

Camera-independent fidelity uses authoritative traffic state and conservation, never visibility, camera location, UI selection, or inspection. The detailed zone contains junction approaches, incidents, stops, and any region where lane choice materially changes flow. Its boundary moves slowly on a schedule derived only from authoritative state. Inspection may promote visual detail and diagnostics but cannot promote simulation fidelity. Outside the detailed zone, road traffic uses lane-group cells and packets; local agents remain individually accountable in a trip ledger.

For a regional cell i , a triangular fundamental diagram yields sending and receiving bounds:

```
S_i = min((v_free_i * dt / L_i) * n_i, Q_i * dt)
R_i = min(Q_i * dt, (w_i * dt / L_i) * (N_jam_i - n_i))
```

```
For one isolated transfer before the node solver:
y_(i->j) = min(S_i * p_ij, R_j)
```

Here n_i and N_{jam_i} are vehicle counts, Q_i is vehicles per transport-second, w_i is backward-wave distance per transport-second, and L_i is cell length. The network node solver allocates all competing movements together, with $y_{ij} \geq 0$, $\sum_j(y_{ij}) \leq S_i$ for every inbound cell, and $\sum_i(y_{ij}) \leq R_j$ for every outbound cell. Deterministic movement priorities resolve oversubscription. Integer transfers use residual accumulators so fractional demand is conserved across ticks. Cell construction enforces the CFL conditions $v_{free_i} * dt \leq L_i$ and $w_i * dt \leq L_i$; otherwise the cell length must increase or the transport timestep must be reduced. Subdividing a cell without reducing dt would worsen this stability constraint.

This follows the conservation structure of Daganzo's cell transmission model ([Daganzo, The Cell Transmission Model: Network Traffic](#)). SUMO likewise distinguishes microscopic and mesoscopic queue simulation and documents the speed/detail tradeoff ([SUMO traffic simulation theory](#), [SUMO mesoscopic model](#)).

Packets are homogeneous only by next portal, vehicle class, occupancy/cargo class, and time bucket. They store count, total people, cargo by commodity, sum of entry ticks, and deterministic ordered member IDs. Expansion reserves physical slots from front to back, then instantiates members in stable order with inherited entry time. Aggregation occurs only after vehicles cross a boundary and meet compatible packet keys.

The transition invariant for every boundary and tick is:

```
people_before = people_after + people_completed
cargo_before = cargo_after + cargo_delivered
vehicles_before = vehicles_after + vehicles_parked_or_retired
occupancy_lane_seconds += physical_occupancy * dt
travel_time(member) = tick_now - original_departure_tick
```

Completions and retirements require destination or lifecycle receipts. A residual ledger carries integer rounding between ticks. A packet blocked by downstream storage remains upstream, so regional queues spill back into microscopic space and vice versa. There is no off-map sink except a modeled gateway with finite admission, queue, and external travel-time contract.

Vehicle dynamics and lane change

Each local vehicle holds route corridor, lane, longitudinal millimetres, speed, acceleration, length, desired headway, and deterministic behavior parameters. A collision-free longitudinal update computes a free-speed candidate and a safe-speed candidate from leader gap and braking bounds, then applies acceleration and jerk clamps:

```
v_(t+1) = max(0, min(v_free, v_safe, v_t + a_max*dt))
x_(t+1) = x_t + v_(t+1)*dt
```

Integer division uses specified rounding, and simultaneous decisions read an immutable tick snapshot before ordered commit. SUMO documents this synchronous previous-step pattern and the safety interactions of car following and lateral movement ([SUMO safety documentation](#)).

Lane changes are finite-duration reservations, not instantaneous swaps. Candidates score route necessity, speed gain, keep-side policy, transit priority, courtesy, and risk. A move is accepted only if front and rear safe-gap forecasts hold throughout the lateral interval and no overlapping reservation has earlier (`request_tick`, `priority_class`, `vehicle_id`). This safety-plus-incentive split is closely related to the published MOBIL model ([Kesting, Treiber, and Helbing, General Lane-Changing Model MOBIL](#)). Mandatory merges gain urgency as the lane end approaches, but never waive physical safety. If a vehicle misses its turn, it continues and replans.

Junctions, signals, queues, and deadlocks

Every junction compiles legal movements and a sparse conflict graph. Unsignalized controllers allocate conflict-zone time intervals using priority, arrival time, gap acceptance, and stable ID. Signal controllers run a deterministic phase state machine: minimum green, extension, yellow, all-red, and optional transit/emergency priority. Detectors are simulation queries over lanes, not rendered trigger volumes.

Vehicles request entry only when they can clear into downstream storage. Let b_m be downstream free length for movement m ; admit vehicle v only if $b_m \geq \text{length}_v + \text{stop_gap}$ and all conflicts are clear. This single rule makes queue spillback physical. A downstream queue can block a turn, then a junction, then upstream junctions. Capacity and demand should be validated against established transport practice rather than tuned solely for appearance; the Transportation Research Board defines capacity around whether a facility can meet demanded flow ([TRB Highway Capacity Manual overview](#)).

Deadlock detection builds a wait-for graph among occupied conflict zones, lane tails, platform blocks, berths, and construction closures. Strongly connected components persisting beyond a **proposed, unvalidated 30 simulated seconds** are diagnosed. Recovery uses legal actions in order: cancel uncommitted reservations; hold upstream inflow; grant one deterministic movement temporary priority; reverse only explicitly reversible lanes; finally mark the network incident and require routing around it. Never delete vehicles. Export the SCC, occupancy, controller phases, and last 256 decisions for reproduction.

4.5 Parking, walking, cycling, and transit

Parking is a capacity network. Each stall or aggregated facility has access lane, pedestrian exit, restrictions, price, and occupancy. A traveller chooses a facility during route planning, but receives only a soft expectation. Near arrival it requests a reservation; if unavailable, it chooses another facility or performs an explicit search loop. Search traffic occupies roads, time accumulates, and the person cannot leave the vehicle until parked. Curb parking competes with loading, bus stops, bike lanes, waste collection, and incident access through time-windowed curb allocations.

Pedestrians route on walkable polygons and connector graphs. Locally, use preferred velocity plus short-range collision avoidance and wall constraints; the classic social-force formulation combines acceleration toward desired velocity with interpersonal and boundary repulsion ([Helbing and Molnar, Social Force Model for Pedestrian Dynamics](#)). For game stability, cap forces and use deterministic neighbor order. Crosswalks are junction movements with conflict and signal control. Crowds queue at finite-width gates, stairs, elevators, fare barriers, platforms, and doors. Regional pedestrians become packets on connectors under the same people/time ledger.

Bicycles use dedicated or shared lane bands, vehicle-like longitudinal occupancy, bicycle-specific acceleration and gap rules, and dismount transitions where required. Accessibility profiles constrain slopes, stairs, elevator use, crossing time, and transfer minimums.

Transit service separates plan from operation:

- `LinePattern`: ordered stops, permitted path, calendar, target headway or timetable;
- `VehicleRun`: assigned vehicle, block of trips, crew abstraction, current delay;
- `StopCall`: platform, scheduled/actual arrival and departure, boarding/alighting ledger;
- `PassengerJourney`: access, transfer, ride, egress legs and fallback alternatives.

Vehicles are ordinary physical agents subject to congestion, signals, block occupancy, depot capacity, and breakdowns. At a stop, alight first, then board by arrival order and eligibility until capacity or departure rule. Dwell time is $\text{base} + \text{board_count}/r_{\text{board}} + \text{alight_count}/r_{\text{alight}}$, with deterministic integer rates. Left-behind passengers remain in a visible queue. Frequency-based lines dispatch to target headway with recovery bounds; timetable lines preserve published slots where feasible and record cancellation explicitly.

4.6 Rail, freight, water, air, and services

Rail uses directed track blocks, switches, signals, platforms, depots, and interlockings. A route request locks a compatible sequence of blocks and switch positions; conflicting routes cannot clear simultaneously. Train dynamics respect length, braking curve, gradient, consist power, platform fit, and turnaround time. A train may enter a block only with authority and a braking envelope contained within granted territory. Junction deadlock logic sees trains, but recovery never teleports or reverses a consist unless the track and operating plan permit it. UIC's RailSystemModel is a useful public reference for scalable railway infrastructure and topology concepts ([UIC RailSystemModel](#)).

Freight is conserved batches with commodity, amount, owner, origin, destination, due window, containerization, and custody state. Terminals transform custody through finite resources: gate, storage slot, crane, dock, siding, worker/equipment abstraction. Trucks, trains, ships, and aircraft carry manifests; loading moves quantity atomically from terminal inventory to manifest at completion. Missed connections create storage and delay rather than disappearance. Empty equipment and return legs are modeled because they consume capacity.

Ships follow waterway centerlines with depth, width, bridge clearance, current, berth, lock, and one-way meeting constraints. Large ships reserve channel and berth intervals; smaller craft use lane-like encounter rules. Water-level changes can invalidate draft clearance and produce a closure transaction.

Aviation is deliberately operational rather than aerodynamic. Airports contain stands, service roads, taxiway segments, holding points, runways, gates, and approach/departure slots. Aircraft reserve stand, taxi segments, and runway windows. Wake-separation class, runway occupancy, weather acceptance, turnaround resources, passenger/cargo transfer, and diversion gateways determine throughput. Airborne cruise outside the map uses a finite external link with travel-time and capacity accounting, while surface movement remains local.

Service dispatch - ambulance, fire, police, road maintenance, hearse, waste, snow, utility repair - uses the same router plus eligibility, equipment, load, station readiness, and urgency. Dispatch solves a bounded assignment: candidate travel time + readiness + coverage penalty + service suitability. Reserve a unit before routing; release on failure. Emergency priority may request signal preemption but still observes physical clearance. Multi-stop collection uses rolling insertion heuristics and reoptimizes only uncommitted stops to prevent route thrashing.

4.7 Construction and closure lifecycle

Construction is simulation state, not an instant topology swap:

- 1 **Plan:** validate geometry, cost, land, utilities, and expected disconnected components.
- 2 **Announce:** publish future restrictions so routes can adapt.
- 3 **Prepare:** place work-zone access and acquire materials/equipment.
- 4 **Close:** reject new entrants; drain or legally evacuate existing occupants.
- 5 **Build:** consume labor, equipment time, and materials by milestones; apply noise and capacity effects.
- 6 **Commission:** run structural/topological validation, then publish the new network version at a tick boundary.
- 7 **Restore:** remove temporary links and update terrain, parcels, addresses, schedules, and maintenance assets.

Demolition follows the same closure protocol. Forced player action can strand agents only with an explicit, logged relocation operation that accounts for people, vehicles, and cargo. Save files persist both the active network and in-flight construction transactions.

4.8 Determinism, telemetry, and acceptance gates

Use a fixed authoritative tick and named lower-frequency phases: movement, controllers, dispatch, route customization, regional flow, and statistics. Parallel jobs may read immutable snapshots. Writes go to command buffers, radix-sorted by (phase, partition_id, stable_id, command_kind), then reduced with specified integer arithmetic. Random choices use counter-based streams keyed by (save_seed, system, entity_id, event_ordinal); worker count cannot alter draws.

Every tick emits or incrementally updates:

- lane flow, density, mean speed, queue tail, lost time, and spillback duration;
- junction movement demand, served flow, phase utilization, blocked-exit events, and conflict denials;
- planning queue wait, expansions, cache validity, repair causes, and route regret;
- passenger wait, denied boarding, dwell, transfer failure, occupancy, and on-time performance;
- freight inventory by custody state, terminal utilization, missed connections, and tonne-hours;
- conservation residuals for people, vehicles, cargo, parking, platform occupancy, and external gateways;
- job time, memory, changed partitions, and worst stable IDs for each budget.

Required automated gates follow.

Replay determinism. Run a scripted 30-day city twice with different render frame rates, cameras, headless mode, worker counts, and job chunk sizes. Hash authoritative SoA pages and ledgers every 1,024 ticks. All hashes and event logs must match. This is an absolute gate.

Edit fuzzing. Property-generate 100,000 **proposed, unvalidated** road/rail/path edits around near-collinear, tangent, tiny-edge, steep-grade, bridge, and tunnel cases. After each accepted transaction, run reciprocal-link, conflict, accessibility, manifold-polygon, and route-reachability validators. Rejected edits must preserve the prior hash.

Conservation. For a scripted evacuation plus freight peak, reconcile every boundary each tick and the whole map daily. Required exact integer residual is zero except explicitly recorded external arrivals/departures and construction transformations. Aggregation followed immediately by expansion must reproduce member IDs, total people/cargo, departure ticks, and occupied length.

Traffic fundamentals. On ring road, merge, signal, gridlock, and lane-drop fixtures, compare flow-density curves, shockwave direction, saturation discharge, queue growth, and spillback with configured targets. The acceptance bands are calibration outputs; no universal values are hard-coded. Collision count and illegal connector use must be zero in nominal fixtures.

Routing validity. Randomly compare hierarchical results with full time-dependent Dijkstra on small snapshots; generalized cost must match within the declared approximation bound. Mutate one partition and prove unrelated cache entries remain valid while every affected path repairs before entering an invalid edge. Inject overload and verify deficit-round-robin service plus zero starvation.

Multimodal operations. Test missed buses, full platforms, blocked rail junctions, port storage exhaustion, unavailable parking, runway closure, inaccessible stations, and emergency preemption. Expected result is explicit queueing, delay, diversion, cancellation, or failure - not disappearance.

Scale. A reference metropolis and hardware profile must be frozen before optimization. The launch qualification tier is a **proposed, unvalidated target** of 250,000 current residents, with active travellers and microscopic movers reported separately from population. The 1,000,000-current-resident city is an experimental stress tier, not the launch contract. A separate **proposed, unvalidated target** permits up to 2,000,000 retained trip-ledger records; those records may include completed trips and historical members and therefore are not 2,000,000 active agents or residents. At normal speed, p95 authoritative tick must remain below its wall-time performance budget, no

subsystem may carry work debt beyond 60 ticks, and memory high-water marks must remain below the platform allocation. Any miss reports the dominant stable IDs and phases rather than changing fidelity according to the camera.

This design makes congestion, missed connections, parking scarcity, freight delay, construction disruption, and service coverage consequences of the same conserved network state. Presentation can simplify meshes and animation freely; the city's transport outcome remains a reproducible result of topology, demand, capacity, policy, and time.

5. Economy, citizens, and services

5.1 Purpose and evidence boundary

This chapter specifies a buildable economy and population simulation for a new city builder. It is a design proposal, not a claim about the implementation of *Cities: Skylines II*. Official material is useful here because it identifies player-facing expectations: citizens have households and lifepaths; employment, education, shopping, health, leisure, and crime affect both travel and the economy; companies transform inputs into products; utilities have networks and capacity; services may be local or imported; land value, rents, accessibility, and pollution affect location choices. The official [Citizen Simulation & Lifepath](#), [Economy & Production](#), [Zones & Signature Buildings](#), [City Services](#), and [Electricity & Water](#) diaries describe those visible behaviors.

They also show a design hazard. Paradox later said its economic safeguards and flows were insufficiently transparent, and its Economy 2.0 changes altered subsidies, service imports, rents, demand, production, and upkeep ([Economy 2.0, part 1](#); [part 2](#)). The lesson is architectural: causality and accounting must be observable from the first implementation, not added after tuning.

Every number below is a proposed starting value or illustrative example. It is unvalidated. Double-entry accounting can make a game internally consistent; it does not make the model empirically realistic.

5.2 Simulation contract

The authoritative simulation is a headless native C++ library. Rendering, UI, audio, and camera state consume immutable snapshots and submit timestamped commands. They never mutate economic state directly. The same save, seed, command stream, and build compatibility version must produce the same state hashes across every supported worker count.

Use a fixed scheduler:

- 20 simulation ticks per nominal wall second at 1x.
- `dt_transport = 0.05` transport_seconds per tick.
- One calendar day equals 1,800 transport seconds.
- `calendar_seconds = transport_seconds * 48`.
- Vehicle speed and path traversal use transport seconds. Fiscal periods, rent dates, school terms, inventory aging, and policy deadlines use calendar time after explicit conversion.
- A 1 km journey at 13.9 m/s takes about 72 transport seconds, which converts to about 58 calendar minutes. That compression is a gameplay choice for the intended map and trip distribution, not a claim about a physical city.
- Aging may use a separately versioned `age_scale`; it must never depend on wall time.
- At 1x, overload slows real execution honestly rather than skipping authoritative ticks. Faster speeds may process more ticks only when capacity exists.

No subsystem may accept an unlabeled scalar called `time`, `rate`, or `duration`. Types such as `TransportSeconds`, `CalendarMinutes`, `MoneyPerCalendarDay`, and `GoodsPerProductionBatch` prevent silent unit errors.

The core simulation phase graph owns global ordering. This chapter registers typed dependencies with that graph: committed topology precedes accessibility and utility solves; delivered utilities, labor, and shipments precede production; settled production and consumption precede ledger close; ledger close precedes forecasts and published read models. An output may feed a declared later core phase or the next tick, but it cannot recursively re-enter an earlier phase. Economy code must not create a second scheduler or local global-tick order.

5.3 Identity, storage, and transactions

All persistent entities receive a never-reused 64-bit stable ID: person, household, dwelling unit, firm, job slot, building, lot, facility, vehicle, shipment, contract, utility node, service request, incident, and ledger account. Dense slot indices may change after compaction; stable IDs do not. External references store (id, generation), while hot loops resolve IDs through a compact lookup table.

Use structure-of-arrays storage by component:

```
PersonCore: household_id[], birth_day[], life_state[], education[], health_q16[]
PersonWork: employer_id[], job_slot_id[], schedule_id[], wage_cents_per_day[]
Household: dwelling_id[], account_id[], size[], vehicles[], move_reason[]
Firm: building_id[], sector[], account_id[], productivity_q16[], status[]
Inventory: owner_id[], good_id[], quantity_milliunits[], reserved_milliunits[]
Facility: building_id[], service_type[], lifecycle[], capacity[], active_workers[]
```

Quantities and money use integers in declared base units. The ledger account is the sole authoritative owner of a cash balance; households, firms, facilities, and UI records store account_id, not a second balance. Any cached balance is a versioned read-model field rebuilt from a committed ledger snapshot. Fixed-point values use named scaling. Floating point may be used in non-authoritative presentation and in deterministic offline content baking, but not for ledger balances or conservation-critical stock.

A transaction is immutable after commit:

```
Transaction {
    transaction_id, tick, calendar_day, type,
    debit_account, credit_account, money_cents,
    good_id, good_quantity_milliunits,
    source_entity, destination_entity, contract_id,
    causal_event_id, policy_version
}
```

Money satisfies $\text{sum}(\text{all account deltas}) = 0$ for every committed transaction batch. Issuance, retirement, grants, imports, and exports still post both sides through explicit mint, sink, or outside-world counteraccounts; no transaction bypasses the invariant. Every physical good satisfies:

$\text{opening_stock} + \text{production} + \text{imports} = \text{closing_stock} + \text{consumption} + \text{exports} + \text{spoilage} + \text{destruction}.$

Reservations prevent double-selling inventory. A purchase creates an order and reserves goods; pickup transfers custody; delivery transfers ownership; cancellation releases the reservation. Money can move through escrow so a destroyed truck does not leave both buyer and seller owning the same value.

5.4 Citizens and households

The household is the budget, housing, vehicle, and relocation decision unit. The person is the education, employment, health, activity, and lifecycle unit. This split prevents a child from independently renting a home while retaining individual trips and outcomes. Dynamic demographic microsimulation commonly models births, deaths, migration, household formation, and dissolution as explicit transitions; Amarin Siripanich and Taha Rashidi describe an alignment example in [A demographic microsimulation model with an integrated household alignment method](#). The proposal borrows that event discipline, not its calibration.

Person lifecycle states include dependent child, student, employed, unemployed, retired, hospitalized, incarcerated, deceased-awaiting-transfer, and removed. Transitions require guards and produce journal events. Birth creates a person and household membership atomically. Partnership or separation changes membership only after checking dwelling capacity and finances. Death terminates work and education contracts immediately, creates a deathcare request, then transfers or closes the person estate after a configurable delay. Immigration and emigration cross explicit boundary accounts and outside-connection queues.

Daily household disposable income is:

$Y_h = \text{wages} + \text{benefits} + \text{transfers} + \text{investment_income} - \text{income_tax} - \text{payroll_tax}.$

Committed expenses are rent or mortgage, utilities, transport passes and fuel, debt service, food minimum, healthcare copays, and policy fees. Discretionary demand uses a bounded budget:

$D_h = \max(0, \text{cash} + \text{expected_income_horizon} - \text{committed_cost_horizon} - \text{reserve_target}).$

Households never spend predicted income directly. Forecasts affect orders; settlement still requires cash or authorized credit. If essentials cannot be paid, arrears accrue by contract. The UI attributes the shortfall to rent, unemployment, travel cost, utility price, illness, or another ledger-backed reason.

Happiness is not a magic productivity multiplier. Store interpretable dimensions: material security, housing fit, health, safety, environment, access, social stability, and leisure. Each dimension has a decaying event history and current exposure. Overall satisfaction can be a weighted bounded mean:

$H_h = 100 * \text{sigmoid}(\sum_i w_i * \text{normalized_factor_i} + \text{trait_bias}).$

Weights vary modestly by life stage and household composition. The UI shows the top positive and negative terms with their source entity and time window. Consequences are specific: low material security raises move search and benefit demand; poor health raises sickness risk; unsafe conditions alter destinations and emigration propensity. Avoid feeding the same happiness number into every system.

Activity plans, jobs, and mobility

Each person owns a rolling activity plan of home, work or school, shopping, healthcare, leisure, and exceptional obligations. Travel demand follows activities rather than spawning arbitrary trips. Activity-based transport research likewise treats travel as derived from scheduled activities and maintains consistent persons, households, and time; see Aleix Bassolas, Jose J. Ramasco, Ricardo Herranz, and Oliva G. Cantu-Ros, [Mobile phone records to feed activity-based travel demand models: MATSim for studying a cordon toll policy in Barcelona](#). This is conceptual evidence, not a demand calibration for a game.

For a destination d and mode m , a proposed generalized cost is:

$C(p, d, m) = a_p * T_{\text{forecast}} + b_p * \text{money_cost} + c_p * \text{walk} + d_p * \text{transfers} + e_p * \text{reliability_penalty} + \text{schedule_penalty}.$

Use integer or fixed-point components and deterministic tie breaking by stable ID. Sample choice from a seeded logit only at explicit replanning events. Once accepted, a plan becomes a contract with a departure window; congestion or incidents may trigger bounded replanning. The official Traffic AI description emphasizes network travel time, money cost, agent preference, and changing conditions in route choice ([Traffic AI](#)); the formula above is a fresh implementation proposal.

Jobs are slots with required education band, sector skill, shift window, workplace, wage, and fill state. Matching runs in spatial buckets, then scores candidates:

$\text{score} = \text{wage_utility} - \text{commute_cost} - \text{skill_gap_penalty} - \text{schedule_conflict} + \text{unemployment_duration_bonus}.$

Use deferred acceptance in bounded regional batches: firms propose to their best candidates; people retain the best feasible offer; unmatched sides continue for a fixed number of rounds. Existing contracts have inertia so jobs do not churn daily. A worker may accept a lower-skill job but contributes at the slot's productivity ceiling. Absence reduces delivered labor hours rather than deleting employment.

5.5 Land, lots, construction, housing, and demand

Road edits generate frontage segments. A deterministic parcelizer produces lots with stable IDs, legal access edges, buildable polygons, zone permissions, utility connection points, and terrain constraints. Zoning grants permission; it does not create a building. Development proposals compete for vacant or redevelopable lots.

Each proposal estimates an unlevered project value over an explicit horizon of H planning periods using discount rate r per period:

$$NPV = -\text{initial_capex} + \sum_{t=1..H} (\text{free_cashflow_t} / (1+r)^t) + \text{terminal_value_H} / (1+r)^H,$$

where $\text{free_cashflow_t} = \text{revenue_t} - \text{operating_cost_t} - \text{tax_t} - \text{maintenance_t} - \text{scheduled_replacement_capex_t}$. Initial capex includes construction and utility connection costs paid at approval or uses their actual scheduled cash flows. Financing viability is assessed separately with loan-to-value, interest, and debt-service coverage tests. Do not subtract financing cost from unlevered cash flows or also include debt service in them; either a project view or an equity view may be implemented, but never both in the same NPV.

A developer starts only if NPV exceeds a tunable threshold and materials, builders, access, and utility reservations are feasible. Construction is a state machine: proposed, permitted, site preparation, foundation, structure, fit-out, commissioning, occupied. Each state consumes labor and goods. The visible model must never appear operational before commissioning; if an intermediate shell is rendered, its status is inspectable.

Dwellings are units inside buildings, not one household per structure. Each unit stores floor area, bedroom class, quality, accessibility, rent contract, occupancy, utility status, and damage. Asking rent updates slowly from operating cost, vacancy, local bid pressure, and policy constraints:

$$\text{rent_next} = \text{clamp}(\text{rent} + k1 * (\text{occupancy} - \text{target_occupancy}) + k2 * \text{cost_change} + k3 * \text{bid_pressure}, \text{legal_bounds}).$$

Household housing utility combines affordability, space per member, commute accessibility, school access, services, environment, safety, and moving cost. Choice sets are sampled by region and price band, then evaluated against network travel forecasts. Land value is an appraisal signal derived from cleared rents, sale proxies, accessibility, environmental burden, and development option value. It is not money and cannot pay bills. Smooth it spatially and temporally, but retain source contributions.

Zone demand is a diagnostic forecast, not a spawning quota:

$$\text{demand_z} = \text{feasible_searchers_z} - \text{suitable_vacancies_z} + \text{expected_firm_expansion_z} - \text{active_pipeline_z}.$$

Show why demand is constrained: insufficient households, unaffordable rents, missing workers, poor freight access, unavailable utilities, high tax, or an already large construction pipeline. Official descriptions connect land value and rent to needs, accessibility, pollution, transport cost, and household or company finances ([Zones & Signature Buildings](#)); this proposal makes those links explicit and auditable.

5.6 Firms, production, inventory, and trade

A firm has accounts, job slots, recipes, machines or service capacity, input and output inventories, supplier contracts, customer orders, and a location. A recipe is integer stoichiometry plus labor and utility requirements:

```
Bakery batch:
  consume 60 flour + 10 energy + 8 labor-hours
  produce 50 food
  create 2 solid-waste
```

Production for batch b is limited by the minimum feasible input ratio, machine capacity, delivered labor, and utility availability. Partial production rounds down in base units; remainders stay in inventory. Efficiency improvements reduce specified inputs or time rather than multiplying outputs from nothing.

Firms reorder when $\text{inventory_position} = \text{on_hand} + \text{inbound} - \text{reserved}$ falls below a reorder point:

$$ROP = \text{forecast_demand_over_lead_time} + \text{safety_stock}.$$

Supplier scoring includes unit price, expected travel time, delivery reliability, tariffs, and contract switching cost. Shipments occupy vehicles or cargo capacity. Imports originate at an outside inventory account and exports

terminate there, so both goods and money cross visible boundary accounts. Agent-based supply-chain research treats inventory, production, and transport as coupled processes with capacity and backorders; Qingqi Long and Wenyu Zhang provide one such framework in [An integrated framework for agent based inventory-production-transportation modeling and distributed simulation of supply chains](#). The game should use a smaller deterministic design suited to interactive latency.

Prices update at market intervals, not each tick:

```
price_next = clamp(price * exp(k * (unfilled_orders - excess_stock) / market_scale),
min, max).
```

Cap movement per interval and publish the causes. Consumer retail prices include acquisition cost, transport, wages, rent, taxes, expected spoilage, and margin. Bankruptcy occurs after sustained insolvency and failed refinancing, not one bad tick. Liquidation cancels orders, pays secured claims in fixed priority, releases jobs and premises, and writes remaining losses to explicit creditor or boundary accounts.

5.7 Municipal finance and explicit support

The city has separate operating, capital, debt, reserve, and restricted-grant accounts. Taxes are assessed from recorded bases and paid on schedules. Service fees correspond to delivered or subscribed service. Capital construction creates an asset and consumes cash; depreciation is optional reporting, while maintenance is an actual expense. Bonds create cash and principal liabilities, then post interest and principal payments by contract.

Subsidies must never silently fill a deficit. Difficulty presets may create explicit grants such as Starter Municipality Grant: 25,000/day until population 2,000, with payer, expiration, and budget-line visibility. Emergency service imports similarly require an enabled policy, price schedule, response delay, and transaction record. The Economy 2.0 diary is useful evidence that invisible safeguards can weaken agency even when intended to help new players ([Economy 2.0, part 1](#)).

Toy one-day ledger

Illustrative values only:

Opening balances:

households cash	20,000
bakery cash	5,000
city cash	10,000
outside-world cash	5,000
bakery flour stock	100 units

Transactions:

bakery imports 60 flour:	bakery cash -600; outside cash +600 outside flour -60; bakery flour +60
bakery payroll:	bakery cash -800; households +800
bakery production:	flour -60; food +50; waste +2
households buy 40 food:	households -1,200; bakery +1,200 bakery food -40; households food +40
sales tax:	bakery -120; city +120
city clinic payroll:	city -500; households +500
explicit starter grant:	outside -300; city +300
household consumption:	household food -36; consumed-food sink +36

Closing money:

households 20,100; bakery 4,680; city 9,920; outside 5,300
total money remains 40,000.

Goods check:

flour: opening 100 + import 60 - production 60 = closing 100
food: opening 0 + production 50 - sold 40 = bakery closing 10;
household received 40 - consumed 36 = household closing 4
total unconsumed food = 14; consumed sink = 36.
waste: opening 0 + production 2 = closing 2.

This deliberately narrow toy audit covers money, flour, food, and waste only. Its bakery step is the simplified transaction $60 \text{ flour} \rightarrow 50 \text{ food} + 2 \text{ waste}$; energy and labor-hour inputs from the fuller recipe above are omitted, and payroll is a money transfer rather than a labor-unit audit. The grant improves playability while remaining visible. The ledger proves conservation inside that stated boundary but says nothing about whether wages, prices, or grant size feel right.

5.8 Utilities and service activation

All facilities share a lifecycle: planned, constructing, commissioning, active, degraded, disconnected, damaged, closing. Appearance, collision, jobs, costs, capacity, dispatch eligibility, and UI must read the same lifecycle record. A building can be visibly under construction without contributing capacity. Once it looks complete but awaits staffing or connection, a visible status names the blocker.

Electricity is a capacitated graph with generation, storage, substations, edges, loads, loss approximation, and priority classes. Resolve islands first, then allocate supply through edge capacities using a deterministic approximate flow. Report unmet load by cause: insufficient generation, disconnected island, saturated edge, damaged asset, or fuel shortage. Track interruption duration and frequency by customer; IEEE 1366 defines distribution reliability indices for circuits and regions ([IEEE 1366](#)). Game metrics may use analogous names only if definitions are shown.

Water and sewer are separate directed capacity graphs. Water balance is source extraction plus treatment plus imports minus demand, leakage, storage change, and exports. Sewer generation derives from delivered water and process coefficients; it cannot exceed its causal inputs. Backups occur at constrained components and propagate upstream. Pollution concentration travels with discharged mass, flow, and decay. The official electricity and water design describes network capacity, trade, groundwater, sewage, and treatment-to-solid-waste coupling ([Electricity & Water](#)); the proposed solver adds strict conservation and causal diagnostics.

Heating is a thermal-demand service driven by building envelope, temperature, occupancy, and process load. It can consume electricity, fuel inventory, or district heat. Avoid counting the same heating demand in multiple networks.

Garbage is a material stock at each building. Collection requests compete for vehicle, crew, route, and destination capacity. Pickup transfers mass to the vehicle; unloading transfers it to landfill, recycling, or incineration. Processing produces recovered goods, emissions, ash, or energy by recipe. Show generation, collected amount, backlog, vehicle capacity, processing capacity, and storage as different quantities.

For every service, distinguish:

- installed capacity;
- staffed and connected capacity;
- scheduled capacity;
- actual delivered service;
- backlog;
- reliability over time.

5.9 Human services, safety, and shocks

Education facilities own seats, teachers, schedules, admissions queues, and progression rules. Enrollment requires a feasible trip or remote option. Learning accumulates from attendance and facility quality; graduation is a threshold event with a seeded assessment, not an instant radius bonus. Education changes job eligibility over time, so new universities cannot immediately solve a current skilled-labor shortage.

Health generates cases from baseline risk, age, pollution exposure, injury events, unmet utilities, and previous conditions. A case progresses through onset, self-care, appointment or ambulance request, travel, triage,

treatment, recovery, and possible death. Facilities have beds, clinicians, diagnostic or treatment capacity, and queues. Passive public-health effects may alter incidence, but must be distinct from treatment throughput.

Police models incidents, reporting delay, dispatch, travel, on-scene work, investigation, arrest, custody, and release. Fire models ignition fuel, detection, spread, dispatch, suppression, rescue, structural damage, and reignition. Neither is solved by colored radius alone. Dispatch selects an eligible unit using predicted arrival plus capability and coverage reserve. The player can inspect why a farther unit was chosen.

Deathcare receives deceased-person custody requests, dispatches a hearse, transfers custody, processes burial or cremation, and closes the request. A body is never duplicated or silently removed. Capacity separates pickup vehicles, storage, and processing.

Disasters are authored or stochastic hazard fields with a recorded seed, footprint, intensity, and timeline. They damage network components and generate ordinary downstream events: injuries, fires, outages, blocked links, displacement, reconstruction demand, and municipal expense. Climate provides calendar-indexed temperature, precipitation, wind, daylight, and hazard modifiers. Official material links weather to energy demand, leisure, road conditions, fire risk, and disasters ([Climate & Seasons](#)); implementation should pass those variables through named inputs rather than global hidden multipliers.

Air, ground, water, and noise pollution have source ledgers. Each emitter records mass or intensity, pollutant type, time, and location. Environmental solvers may be coarse grids, but exposure integrates at a household, workplace, and activity location. Health and land appraisal consume exposure histories. Policies change specific emission factors, operating costs, or enforcement rates.

5.10 Multi-fidelity simulation without identity loss

Camera distance must not determine authoritative existence or simulation policy. Every person, firm, shipment, vehicle, and request keeps its stable ID off screen. Following the transport chapter's policy, fidelity is selected by workload and the declared accuracy contract, independent of the camera:

- Microscopic: individual vehicle movement and explicit queues.
- Mesoscopic: lane or link packets with preserved members and scheduled exits.
- Event driven: origin-destination cohorts or shipment batches carrying member IDs, counts, reserved capacity, money, goods, and expected arrival distributions.

Refinement expands the same cohort into its members; aggregation packs them deterministically. Totals before and after must match. A camera jump requests rendering detail only and cannot change the simulation representation, employment, inventory, service response, or travel time. Statistical background updates are permitted for low-impact decisions only when their error budget is bounded and they emit the same kinds of transactions as detailed updates.

Partition work by deterministic regions and component type. Workers write command buffers; a serial or ordered merge commits by (phase, region_id, entity_id, event_sequence). Cache accessibility matrices and route trees by network version. Stagger expensive household moves, firm searches, appraisals, and market clearing across deterministic cohorts.

5.11 Player causality and diagnostics

Every visible aggregate supports drilldown:

```
city total -> district -> building/facility -> entity/request -> causal events -> transactions/routes.
```

A service panel shows installed, active, dispatched, delivered, backlog, imports, price, and reliability. A building explains No garbage pickup: road access disconnected at node 1842, not just a red icon. Demand bars expose positive and negative terms. A household budget lists actual ledger categories. A company panel shows recipe bottlenecks, input days of supply, unfilled orders, labor shortage by skill, utility curtailment, and recent

profit.

Keep a bounded history ring for each entity and a longer aggregated event journal. Debug builds can retain full transaction and event logs. State hashes exist per subsystem and region so a divergence identifies the first failing slice. Breakpoint-free instrumentation includes sampled traces, event counters, queue histograms, route versions, transaction reconciliation, and facility lifecycle transitions.

Dependency graph and policy application

Treat the following as the principal causal graph. Arrows mean an output becomes a named input at a later phase or tick:

```
climate/weather -> utility load, hazard risk, leisure choice, crop yield
land use + networks -> accessibility -> household/firm location bids
households -> labor supply + consumption orders + activity travel
firms -> jobs + goods + freight + wages + taxes + emissions
utilities -> delivered service -> health, production, household costs
education -> qualifications -> job matching -> wages/productivity
pollution + safety + material security -> health/well-being -> behavior
incidents -> dispatch + travel -> service outcome -> damage/backlog
all settled flows -> ledgers/stocks -> forecasts -> next decisions
```

Cycles cross a commit boundary. For example, a new factory may create jobs during today's firm phase, but households reconsider employment on the next matching interval and housing on the next relocation interval. This prevents same-tick positive feedback from making a factory instantly fill, pay wages, create demand, raise rent, and level its building.

Policies are versioned parameter bundles with scope, start tick, end tick, eligibility predicate, funding account, and explicit effects. Effects modify named terms such as `industrial_emission_factor`, `transit_fare`, `property_tax_rate`, or `clinic_copay`; scripts cannot write arbitrary entity state. Enactment emits a policy event, invalidates only affected caches, and records the version on resulting transactions. Forecast panels show baseline, policy delta, confidence range, and earliest plausible response. A congestion charge can alter route and mode choice quickly, while an education subsidy should affect enrollment before qualifications and wages.

Difficulty presets select visible initial conditions and parameter sets: starting debt, grant contract, outside-service prices, disaster frequency, credit terms, and advisory assistance. They do not grant hidden production, erase losses, or change conservation. Balance tests compare failure and recovery paths: how long a solvent city can absorb a revenue shock, whether a beginner preset exposes the cause before insolvency, and whether an expert preset demands planning without requiring foreknowledge of hidden rules.

5.12 Scenarios, invariants, and performance gates

Build tiny deterministic scenarios before a full city:

- 1 One household, one dwelling, one employer, one shop: verify wages, commute, purchase, consumption, rent, and tax.
- 2 One bakery and one imported input: delay a shipment, observe inventory depletion, production stop, price response, and recovery.
- 3 Two neighborhoods behind one constrained power edge: confirm curtailment, cause text, and interruption metrics.
- 4 A newly built landfill without road access, then connected: lifecycle, dispatch eligibility, garbage custody, and panel totals must agree.
- 5 One clinic with two simultaneous cases: queue order, travel, treatment, absence, billing, and outcomes.
- 6 Fire during congestion: route selection, arrival forecast error, spread, damage, injury, and recovery expense.
- 7 Flood or tornado cutting a utility and road: no lost identities or duplicated shipments across fidelity changes.

- 8 Housing shock: close a major employer, trace arrears, cheaper-home search, vacancy, rent adjustment, benefits, and migration.

Hard assertions run continuously: no negative unreserved inventory; no person in two households; no job held twice; no occupied dwelling beyond permitted capacity unless marked overcrowded; no service vehicle assigned to two requests; money and goods reconcile; every active facility is commissioned, connected, and staffed; every displayed total has an authoritative source tick.

Performance telemetry reports time and bytes per subsystem, entities processed, cache hit rate, path requests, market candidates, events emitted, and worst queue length. Gates should use representative saves and the core chapter's fixed hardware target. At 20 Hz, 50 ms is a hard tick ceiling; plan for 35 ms nominal total work to retain recovery headroom. This chapter does not assign competing subsystem budgets. Track p50, p95, p99, and the worst ten-second window; averages hide spikes.

5.13 Tradeoffs and evidence still needed

Full individual accounting provides strong stories and debugging but costs memory and makes aggregate calibration harder. Multi-fidelity transport reduces cost while creating promotion, demotion, and arrival-distribution complexity. Integer conservation prevents value creation bugs but requires deliberate rounding accounts. Slow price and location updates improve stability while delaying feedback. Explicit service queues make failures legible but can frustrate players unless staffing, access, and response forecasts are visible. Subsidies and imports improve recoverability, but only when explicit enough that players can understand the counterfactual budget.

Before tuning, collect distributions rather than single targets: household sizes, trip lengths, commute modes, vacancy duration, rent burden, job matching time, firm inventory cover, service utilization, response time, outage duration, municipal revenue mix, pollution exposure, and disaster losses. Decide which are modeled for recognizable causal gameplay and which are simplified. Calibrate small scenarios first, then test cross-system shocks. Preserve every failing seed and command stream as a regression case.

The core success criterion is not maximum detail. It is that the city reacts through conserved flows, stable identities, explicit contracts, and inspectable causes - and that the same mechanisms still function when the camera is elsewhere and the city is large.

6. Presentation, content, interaction, and modding

6.1 Architectural boundary and engine choice

The presentation product is a client of the native C++ simulation, not its owner. The simulation runs headless, advances authoritative state, validates commands, and publishes immutable render snapshots. The presentation process may interpolate snapshots, predict cursor feedback, and discard work under load; it may never determine money, travel, service coverage, land value, path choice, or any other outcome. Camera position, visibility, frame rate, and graphical settings therefore cannot change simulation results.

Unreal Engine is the provisional presentation adapter. A production spike must earn that choice against a thin custom renderer on representative cities, build size, iteration speed, platform support, frame pacing, memory, and integration cost. The production spike uses the agreed reference scale of 50,000 buildings and 20,000 moving agents, along with large road networks, rapid camera movement, nighttime weather, and a content hot reload. An optional synthetic test may push one million lightweight presentation instances to expose batching and culling limits, but it is neither a production-scale requirement nor a substitute for the representative-city spike. The spike must not substitute Actor-heavy prototypes for the proposed architecture.

Epic describes MassEntity as a data-oriented gameplay framework, while World Partition is an Actor and grid-cell streaming system ([MassEntity](#), [World Partition](#)). Both may help implement an adapter, but neither is part of the simulation contract. MassEntity and any other engine subsystem whose support status changes by Unreal version must sit behind project-owned interfaces; no save format, mod API, or authoritative identifier may contain an Unreal class path. World Partition may stream authored scenery, while simulation-derived roads, buildings, vehicles, people, and overlays use project-owned spatial pages.

The baseline boundary is a versioned, typed C++ snapshot API within the process. Simulation storage is never exposed directly: the simulation publishes immutable snapshot objects through project-owned interfaces, and presentation retains only reference-counted read handles. If process isolation or an external headless viewer later justifies the complexity, the same logical schema may gain a separately versioned serialized representation over shared memory or a local transport. Remote transport is optional and is not required for the shipping desktop architecture. A snapshot is logically:

```
RenderSnapshot {
    protocol_version, tick, sim_time, world_origin_cm,
    changed_page_ids[], removed_page_ids[],
    entity_pages[], network_pages[], terrain_pages[], overlay_pages[],
    event_batch, string_table_delta, checksum
}

EntityPage {
    page_id, generation, bounds,
    stable_ids[N], archetype_ids[N], positions_q[N], rotations_q[N],
    animation_state[N], tint_or_variant[N], flags[N]
}
```

Arrays are structure-of-arrays, compact, and bulk-copyable. Positions are quantized relative to the page; stable IDs survive culling and reload. Changed pages are double or triple buffered and published atomically. The renderer holds snapshot T and $T+1$, interpolates transforms, and treats teleports and topology changes as discontinuities. Camera-relative transforms keep GPU precision local even if the authoritative world uses 64-bit coordinates. Unreal documents its Large World Coordinates as double-precision core types, but also notes that Niagara uses float-based tiled positions for performance, reinforcing the need to test every presentation subsystem rather than assume identical precision ([Epic LWC](#), [Niagara LWC](#)).

No citizen becomes an Actor. Stable groups are uploaded to GPU instance buffers; exceptional selected entities get temporary presentation proxies. A proxy contains only selection, animation, audio, and inspection state and is destroyed without affecting the simulation. Similar rules apply to vehicles, trees, street furniture, and building props.

Proposed, unvalidated spike gates: a snapshot publish under 4 ms at the 99th percentile; presentation ingest under 3 ms; no simulation-thread wait on rendering; under 64 bytes of live presentation metadata per ordinary citizen; and identical authoritative checksums across camera scripts and graphics presets. All numbers in this chapter are proposals until hardware captures validate them.

6.2 Rendering strategy and measurable quality

The default path is scalable deferred or clustered raster rendering. Hardware ray tracing is an optional quality layer for selected reflections, shadows, or global illumination, never a minimum content assumption. Epic's current hardware ray tracing documentation lists partial geometry support and notes that Nanite meshes may use fallback representations, which can create both cost and mismatch ([Epic hardware ray tracing](#)). Every ray feature therefore needs a raster fallback and a separate capture suite.

Nanite is a candidate for opaque, mostly rigid architecture and terrain details after measurement. It does not eliminate disk, streaming, material, translucency, or silhouette costs. Epic explicitly warns that source detail still affects package size and describes fallback meshes for cases that cannot use the Nanite representation ([Epic Nanite technical details](#)). Foliage, animated vehicles, crowds, decals, wires, glass, road markings, and user assets retain explicit conventional LOD paths until the spike proves a better choice. Shipping must not depend on a feature marked Experimental or Beta in the pinned engine version.

Frame and screen-space budgets

Budgets are defined per quality tier and view class: street, neighborhood, city, and map. At 60 Hz, CPU and GPU work overlap rather than summing into one serial allocation. The proposed presentation targets are at most 6 ms of CPU critical-path work and at most 13 ms of GPU work inside the 16.67 ms frame deadline, leaving platform-specific latency and variance margins that must be measured end to end. The authoritative simulation runs separately at 20 Hz, with a proposed 35 ms target and 50 ms ceiling across its worker jobs; presentation interpolation bridges its ticks. A 30 Hz presentation tier receives its own budgets rather than silently running a missed 60 Hz target. Track median, 95th, 99th, and worst CPU critical path, GPU frame, simulation tick, and end-to-end frame times; average FPS cannot approve a feature.

Every family has a screen-space contract:

- Trees preserve crown shape and trunk mass before leaf detail. At distance, transition from masked leaf clusters to low-frequency cards or impostors; never flatten an entire crown into a visibly planar cross. LOD changes must remain below a proposed 1.5-pixel silhouette error and a proposed 3% projected-area jump.
- Crowds preserve motion direction, body height, and density. Distant people become batched animated impostors or density fields, with no individual skeleton evaluation outside its useful pixel size.
- Vehicles preserve roofline, wheelbase, color block, lights, and lane-relative motion. Wheel and interior animation drop before body silhouette. Emergency and transit identifiers survive longer than cosmetic trim.
- Buildings preserve skyline, setback, roof profile, and major facade rhythm. Interior parallax and small props have explicit distance and nighttime cost limits.

Those thresholds are starting hypotheses. Automated orbit captures calculate silhouette difference, temporal edge energy, overdraw, visible instance count, triangles, material slots, texture residency, shadow pages, animation time, and GPU milliseconds. The asset dashboard rejects a tier that saves triangles but produces a larger temporal pop or shader cost.

Temporal stability and reconstruction

The city camera creates hard cases for temporal reconstruction: thin wires, grilles, leaf masks, road markings, specular windows, particles, fast zooms, and huge fields of repeated geometry. The adapter supports native-resolution TAA, Unreal TSR, and platform upscalers through one configuration layer. Epic explains that TSR reconstructs from current and prior frames and that temporal accumulation has limits for thin or moving detail; its higher-quality history can also carry significant GPU cost ([Epic TSR](#)). We therefore evaluate motion clarity, ghost trails, disocclusion, foliage shimmer, and text/overlay sharpness, not only still screenshots.

All moving instances output valid velocity, including vertex animation and impostor transitions. Selection outlines and analytic overlays render after temporal upscaling or provide correct responsive masks. History resets on camera cuts, origin shifts, time-of-day jumps, topology swaps, and large exposure discontinuities. Dynamic resolution has hysteresis and a minimum dwell time so a borderline frame does not create visible oscillation. A sharpening pass is capped to prevent ringing around roads and labels.

Proposed acceptance: on scripted pan, orbit, and zoom routes, no persistent trail longer than two frames behind a vehicle; no one-pixel road marking disappears for more than one frame; LOD transitions do not create a luminance step above 5% over a controlled background; and 99th-percentile presentation frame time stays within its tier. These image thresholds require a perceptual calibration pass before becoming gates.

Lighting, exposure, night, and effects

Lighting communicates time, safety, land use, and service state without turning the city into an exposure demo. Use a physically coherent sun/sky baseline, bounded local lights, and a deterministic time-of-day parameter from the snapshot. Auto exposure uses separate indoor/editor and outdoor gameplay profiles; gameplay exposure has proposed adaptation limits and min/max EV ranges per weather state. Selection colors and service overlays are evaluated after tonemapping so their meaning survives noon glare and night darkness.

Bloom has a luminance threshold, energy clamp, and maximum screen radius. Rain, snow, steam, fireworks, headlights, and decorative particles each have a screen-area allowance, spawn priority, and overdraw ceiling. Night does not multiply every emissive and particle system. When a budget is exceeded, decorative emitters reduce rate first, then lifetime and resolution; safety signals, selection, and transport indicators retain priority. The GPU capture reports translucent pixels drawn per output pixel, particle simulation time, light count by type, and bloom contribution.

This applies lessons from the team's Slopmerica reference review: avoid flat foliage, implausibly dense boats, excessive nighttime particles and bloom, a HUD that consumes the view, clipped tooltips, and ambiguous service activation. These are visual lessons from that separate reference, not claims about Cities: Skylines II.

Terrain, water, and weather

Terrain is authored and streamed in height/material pages, with deterministic edit patches owned by the simulation/save layer. Render meshes add skirts or shared edge samples to prevent cracks. Material layers are slope-, height-, climate-, and biome-aware, but macro variation is baked or procedurally seeded so distant terrain does not become noisy. Collision and hydrology never read displaced render-only vertices.

Water rendering consumes authoritative surface height, flow, contamination, and shoreline masks. It may add bounded visual waves, foam, refraction, and reflections, but a visual wave cannot flood a cell or move a boat. Reflections degrade from optional rays to planar or screen-space approximations to probes. Weather uses one authoritative climate state plus local presentation volumes. Wetness, snow cover, wind response, fog, and cloud shadow fade with stable time constants; rapid camera movement must not reveal empty weather cells. Boats are instanced according to actual transport and leisure state, with aesthetic filler capped by an explicit density field and disabled when it obscures readable traffic.

6.3 Roads, intersections, buildings, and the content pipeline

Road visuals consume the canonical surface and boundary recipe emitted by the authoritative network compiler as part of a validated network transaction. The renderer receives stable node, segment, lane, sidewalk, marking, surface, elevation, and state identifiers plus canonical sampled boundaries, junction regions, layer ordering, and seam keys. It tessellates and shades that result but does not independently regenerate junction topology from centerlines. This keeps routing, placement, collision, save state, and visible intersections on one accepted topology. The authoritative compiler must handle acute angles, grade changes, medians, tram or rail crossings, tunnels, retaining walls, crosswalks, and left/right-hand traffic; presentation validation then catches tessellation defects such as z-fighting or inverted triangles.

```
RoadVisualRecipe {
    recipe_guid, schema_version,
    network_transaction_id, canonical_surface_regions[], boundary_loops[],
    layer_order[], seam_keys[],
    compatible_lane_roles[], width_rules[],
    surface_material, marking_set, curb_profile, sidewalk_profile,
    prop_rules[], bridge_set, tunnel_set,
    lod_policy, accessibility_tags
}
```

Road meshes are tessellated from accepted canonical regions per dirty spatial page, not reconstructed per segment per frame. Markings use decals or mesh strips selected by capture evidence; intersection markings are clipped against the canonical 2D regions. A geometry validator checks transaction and seam-key agreement, finite coordinates, winding, degenerate triangles, self-intersections, non-manifold edges where prohibited, UV density, tangent validity, bounds, collision policy, material count, and seam continuity. Golden topology cases render from above and street level, including edit previews and undo/redo.

The source exchange format for ordinary static assets is glTF 2.0 plus project extensions; the Khronos specification defines glTF as an API-neutral runtime delivery format with structured scenes, meshes, materials, skins, and animation ([Khronos glTF 2.0 specification](#)). Source files are never loaded directly into a live city. A deterministic cooker resolves units and axes, validates data, generates collision and conventional LODs, optionally builds Nanite data, compresses textures, computes derived bounds and costs, and emits a signed manifest plus platform payloads.

```
AssetManifest {
    asset_guid, source_hash, cooker_version, schema_version,
    semantic_type, dependencies[{guid, version_range}],
    bounds, sockets[], variants[], localization_keys[],
    render_costs[{lods[], materials, texture_bytes_by_mip, animation_bytes}],
    gameplay_descriptor_hash, payload_hashes_by_platform
}
```

Gameplay descriptors are small, engine-neutral records for footprint, zoning class, capacity references, access points, utility connectors, and upgrade slots. Saves reference the stable asset GUID, a semantic type, and the last known descriptor version. Presentation payloads can be recooked without changing simulation meaning. If a cosmetic payload is absent, the client uses a typed placeholder with the correct footprint and connector markers.

Asset validation has severity levels: corruption or unsafe values fail; likely visual defects quarantine the asset from publishing; high cost produces a waiver request with captured evidence. Proposed per-asset checks include maximum material slots by category, maximum texture bytes by tier, projected triangle density, LOD silhouette error, shader instruction class, transparency coverage, skeleton bone count, animation compression error, audio peak and loudness range, and localized-name completeness. These are catalog targets, not universal hardcoded limits.

Streaming, memory, and hitch prevention

Spatial pages have independent CPU geometry, GPU buffers, texture mip requests, collision, audio, and UI-label residency. A camera-motion predictor requests a wide low-detail ring before fine assets. Eviction uses priority, distance, last use, reconstruction cost, and pinned selection state. The HUD exposes separate budgets for CPU committed memory, GPU resources, streaming pool, transient render targets, audio, and mod content.

Degradation follows a deterministic ladder: drop distant animation, higher mips, optional interiors, shadow detail, reflections, and decorative effects before losing road topology or selected-object clarity.

Shader permutations are deliberately small. Materials are built from a bounded feature matrix, and the cooker emits every required pipeline state key. Epic's PSO documentation says precaching compiles asynchronously but consumes threads and memory, and recommends testing with a cleared driver cache because existing caches can hide first-run hitches ([Epic PSO precaching](#)). Test cold launch, first visit to every biome and weather state, all overlay modes, selection, demolition, and day/night transitions on each graphics API. Proposed gate: no presentation hitch over 50 ms and fewer than one frame over 33 ms per ten-minute cold-cache route, subject to platform-specific calibration.

6.4 Camera, selection, UI, and state legibility

The camera is double-precision at its boundary and camera-relative inside rendering. It provides orbit, pan, zoom-to-cursor, tilt, follow, cinematic, and orthographic map modes through data-driven control profiles. Perspective FOV is stored vertically and converted by aspect ratio; user settings expose a safe range and preview. Zoom changes altitude and lens together along a tuned curve so street inspection feels spatial while city scale remains readable. Collision avoidance eases rather than snaps, can be disabled for free camera, and never affects simulation outcomes.

Selection is stable-ID based. CPU broad phase queries snapshot page bounds; GPU or compact CPU fine tests return candidates; a policy then ranks icon, road, building, vehicle, and citizen hits by visibility, depth, cursor distance, and current tool. Repeated click cycles candidates. The hover result shows immediately as speculative UI, while the next authoritative reply confirms command eligibility. Demolition, road edits, and service placement preview all affected cells, cost, conflicts, and expected state transition before commitment.

The HUD is composed as dockable regions around a protected world viewport. At the default layout, proposed persistent UI coverage is under 18% of pixels at 16:9 and under 24% at narrower supported aspect ratios; tools may temporarily exceed that with a one-action collapse. Panels reflow at 16:10, ultrawide, and large text scales. Tooltips are clamped to the safe rectangle, choose an alternate side before covering the target, scroll when necessary, and are tested with longest localized strings. No tooltip may clip or extend beyond the safe area.

UI reads a single `ViewModelFrame` derived from one simulation tick, preventing mixed-time panels. State-bearing controls use explicit states such as `available`, `preview`, `pending`, `active`, `degraded`, `blocked`, and `failed`. A service button never communicates activation by color alone: it combines label, icon, shape or pattern, status text, and an optional sound. Overlays have a legend, units, time stamp, unavailable-data pattern, and comparison baseline. A paused or stale overlay says so.

```
ViewState {
    snapshot_tick, selected_ids[], active_tool, pending_command_ids[],
    overlay{id, metric, units, scale, timestamp, stale_reason},
    panels[{id, anchor, size, focus_path}],
    accessibility_profile
}
```

Unreal Common UI can supply layered input routing and cardinal navigation, but Epic's own guidance says it should be adopted based on actual complex-menu and cross-platform needs ([Common UI overview](#), [Common UI design guidance](#)). We keep a project-owned semantic widget model and focus graph so presentation can migrate and automated accessibility tests can inspect roles, names, values, states, bounds, and traversal.

6.5 Animation and audio

Animation is importance scheduled. Near selected citizens use compressed skeletal clips, motion matching only if its visual gain survives a cost review, inverse kinematics for terrain contact, and phase-preserving state transitions. Mid-distance crowds share pose palettes or GPU animation; far crowds use impostors or flow visualization. Vehicles animate wheels, steering, suspension, doors, and articulated sections by screen contribution. A discontinuity flag prevents interpolation across despawn, teleport, lane remap, or network rebuild.

Audio is driven by snapshot state and discrete events, spatially clustered before it reaches the engine. A block with 600 cars produces a bounded traffic bed plus a small number of salient vehicles, not 600 full voices. Similar aggregation applies to crowds, industry, construction, weather, and wildlife. Priority combines semantic value, distance, visibility, player focus, and repetition. Loops have seeded variation and crossfade across page boundaries; event IDs suppress duplicates when snapshots are replayed.

Use submixes for UI, traffic, people, nature, weather, construction, and music, with user controls and accessibility routing. MetaSounds may implement procedural layers because Epic documents sample-accurate DSP graphs and integration with gameplay parameters, but authored logic remains behind project events and parameter contracts ([Epic MetaSounds](#)). Proposed budgets include 128 rendered voices, 32 high-detail spatial sources, and 2 ms audio-thread time at the reference tier. Validate rather than universalize them.

Acceptance captures measure voice steals by category, event-to-sound latency, clipping, loudness, repetition, page-boundary discontinuities, and whether key UI and service-state cues remain distinguishable with music muted, effects reduced, or mono output.

6.6 Accessibility and localization as system contracts

Accessibility begins in schemas and input routing. Every action is remappable, supports mouse/keyboard and controller, and avoids required holds or repeated rapid input where an alternative can exist. Camera shake, motion blur, depth of field, flashing, weather density, bloom, and UI animation are independently reducible. Time controls let a player pause while reading. Critical information has at least two channels among text, color, icon, pattern, sound, and haptic feedback.

UI components expose semantic name, role, value, state, hint, order, and live-update priority. Narration announces committed changes and actionable errors, not every simulation tick. Charts and overlays provide focusable summaries and inspectable values. Microsoft recommends that core menus, HUD state changes, and time-based events support screen narration, and its guidelines set concrete contrast and subtitle practices ([XAG 106 screen narration](#), [XAG 102 contrast](#), [XAG 104 captions](#)). Treat these as design inputs and test cases, with platform certification requirements tracked separately.

Proposed UI gates include 4.5:1 contrast for ordinary meaningful text, 3:1 for large text and non-text indicators, a 200% text scale pass, no color-only state, complete keyboard/controller reachability, visible focus, and narration of all task-critical state. Caption controls include size, speaker, sound description, background color/opacity, and safe-area preview. Automated tests crawl focus graphs and detect clipping at supported resolution, aspect, language, and scale combinations; disabled players participate in milestone studies.

All player-facing strings use stable localization keys and ICU-aware formatting for plurals, numbers, currency, dates, sorting, and bidirectional text. Unreal's localization layer itself uses ICU and separates gathered source, culture archives, and compiled resources ([Epic localization overview](#)). The project wraps it so simulation events carry keys plus typed arguments, never preformatted English. Pseudolocalization expands text, adds diacritics, and exercises right-to-left layout. Fonts have declared glyph coverage and fallback; UI reserves space based on tested expansion rather than English width.

6.7 Save-safe editing, UGC, and the mod API

Content edits are transactions over stable GUIDs. The editor stages changes in a scratch package, validates and cooks them, then atomically publishes a new immutable asset version. A running city may hot-reload purely visual payloads. Changes to footprints, connectors, capacity semantics, networks, or simulation code require an explicit migration and a save checkpoint. Undo stores inverse operations or prior immutable versions; it does not mutate a file that an existing save silently depends on.

Each save records simulation schema version, mod lockfile, asset GUID and semantic-version requirements, content hashes, granted capabilities, and migration history. Load resolves exact compatible versions before simulation starts. Missing cosmetics receive placeholders. Missing gameplay definitions open a repair report and may load only through a declared fallback or migration; the engine never guesses a new capacity or silently deletes a building. Removing a mod runs its uninstall validator against a copy of the save and presents impacted entities before commitment.

The public API is capability based and split into data, deterministic simulation extensions, presentation, UI, importers, and tooling. Declarative JSON or FlatBuffers-like schemas handle ordinary assets and policies. Code mods compile to WebAssembly components and begin with no filesystem, network, clock, process, clipboard, or arbitrary engine access. WASI is designed around capability-based security with no ambient authority, while Wasmtime documents memory isolation and capability-scoped filesystem access ([WASI design principles](#), [Wasmtime security](#)). This supports the security model, but it is not a complete security proof: pin and patch the runtime, fuzz host calls, validate handles, and isolate high-risk cookers in separate processes.

```
ModManifest {
  mod_guid, version, api_range, content_hash,
  dependencies[{guid, version_range, required}],
  capabilities[read_snapshot, add_asset, register_ui_panel, ...],
  deterministic_modules[], presentation_modules[],
  migrations[{from_schema, to_schema, module_hash}],
  limits{memory_bytes, fuel_per_tick, events_per_tick}
}
```

Deterministic modules receive tick, seeded random streams, immutable query views, and transactional command writers; they cannot read wall time or camera state. Proposed writes stay in a private buffer until the module completes and all commands validate. A trap, quota breach, or validation failure discards the entire tick's proposed writes for that module, records diagnostics, and applies the deterministic failure policy stored with the save; partial writes never enter authoritative state. Execution has fuel, memory, recursion, allocation, output, and event quotas. Presentation modules may be nondeterministic but cannot write authoritative state. Native plugins are a separate developer mode with prominent trust and save-compatibility labeling.

API compatibility uses semantic versions plus feature negotiation. Additive fields have defaults; removed behavior passes through a published, bounded deprecation window; unknown fields round-trip where possible. At launch, contract tests cover every API version that has shipped. As the API history grows, the project may introduce a bounded support horizon only through that published policy and explicit migration tooling. The SDK ships schemas, generated bindings, validators, a headless simulation runner, deterministic replay, performance traces, and packaging/signing tools. Workshop policy and discovery are service concerns layered above local package validity.

Proposed release gates: if the core simulation contract promises bit-exact determinism, deterministic mods reproduce authoritative hashes across supported CPU models for the same build; cross-platform equivalence is a separate gate with its own declared tolerance or determinism contract. A hostile corpus cannot escape granted capabilities; quota exhaustion discards the module's complete write transaction and produces a bounded failure; uninstall dry runs enumerate all affected save records; every API version shipped to date passes contract tests until the published support horizon takes effect; and a presentation-only mod cannot alter an authoritative checksum. These gates, like the numerical rendering and audio budgets above, remain unvalidated proposals until representative content, hardware, and creator studies establish their final values.

7. Persistence is part of the simulation

A save captures a legal tick boundary, including work that has not finished. The city is more than buildings and bank balances. Persist route continuations, pending orders, escrow, inventory reservations, service assignments, construction phases, traffic-controller phases, transit runs, utility storage, pollution fields, event queues, random counters, policy versions, and the conversion between clocks. Anything that can change the next decision is authoritative or must have a documented deterministic reconstruction.

The container has a small manifest followed by independently compressed typed chunks. Each chunk declares type, schema version, entity/tile range, encoded length, decoded length, checksum, and dependencies. The manifest records game build, simulation compatibility version, world seed, tick, content hashes, mod lockfile, ruleset, map coordinate convention, and migration history. Thumbnail, description, and player camera are separate optional chunks. Limits on chunk lengths and decompression ratios prevent corrupt files from allocating arbitrary memory.

```
SaveManifest
  format_version, simulation_version, tick, seed
  clock_config, ruleset_hash, content_manifest_hash
  mod_lockfile, chunk_index[], checkpoint_parent
  authoritative_root_hash, required_features[]
```

```
Chunk
  kind, schema_version, object_range
  compressed_size, uncompressed_size, checksum
  dependency_ids[], bytes
```

A save request pins a consistent snapshot. Background compression and writing operate on immutable pages; subsequent writes clone only modified pages. This copy-on-write path needs a strict extra-memory budget and backpressure. If dirty pages exceed the reserve, briefly pause at a tick boundary or postpone nonessential work; do not risk exhausting memory to avoid a visible save pause. An unbounded snapshot copy can be worse than a short, predictable pause.

Write a new temporary container, flush according to the target platform's durability API, read back or verify the encoded index/checksums, then replace the current save using a tested platform-specific atomic publication procedure. Retain the previous validated checkpoint. Atomic rename is not the same thing as guaranteed power-loss durability on every filesystem; test the actual platform and storage behavior. SQLite's documentation illustrates the distinction between a correct logical commit protocol and assumptions about caches, writes, and failure. We borrow that failure discipline, not SQLite's database layout. [SQLite atomic commit](#).

Autosaves rotate several known-good checkpoints and never overwrite all recovery points after a migration. Cloud synchronization is optional and acts on closed containers. Detect divergent histories by checkpoint ancestry/hash; preserve both versions and let the player choose. Last modification time alone does not establish which world is the intended continuation. Offline play does not depend on cloud availability.

7.1 Load, migration, and replay

Loading validates container bounds, checksums, manifests, dependencies, and required features before constructing a world. Restore authoritative records, rebuild permitted caches in canonical order, validate cross-references and conservation, then publish the first playable snapshot. A load error names the failing chunk or dependency and leaves the original untouched. Missing visual assets can use typed placeholders. Missing gameplay definitions require an explicit compatible substitution or repair workflow.

Migrations are pure versioned transforms from an old schema to a new schema, executed on a copy. They produce counts of changed/removed/introduced records and a reconciliation report. A capacity reduction cannot silently discard excess inventory; it creates a declared overflow or relocation state. A changed road definition cannot silently strand vehicle handles; it supplies a connector remap or a suspended edit requiring repair. Keep fixtures from every released save version and test sequential migrations as well as supported direct paths.

Command replay starts from a checkpoint and feeds canonical commands at their recorded logical ticks. It records the exact executable/rules/content identity. Store hashes at subsystem and regional boundaries so a mismatch can be bisected to one tick and owner. Full byte-for-byte snapshots are useful for debugging but do not replace canonical state hashing; padding, allocation order, and reconstructed caches are excluded. Factorio's save/load discussion explains why preserving deterministic continuation is a stringent requirement. [Factorio save/load overview](#).

Undo has two meanings. In paused planning, inverse edits can restore a prior transaction if dependencies have not changed. Once time advances and residents act on a built road, undo cannot simply refund cash and erase consequences. Offer a preview cancel before commitment, a normal demolition transaction afterward, and optional checkpoint rewind for sandbox experimentation. The UI names which operation it is performing.

8. Environment and coupled infrastructure solvers

The environment needs enough physical structure to make placement matter, while remaining affordable and interpretable. Use tiled coarse fields with optional local refinement around developed land and infrastructure. The authoritative topology determines which cells exchange water or pollution; render-only waves, vegetation motion, smoke, and reflections never feed back into those fields.

For hydrology, store cell ground elevation, water volume, surface level, infiltration capacity, saturation, and boundary conditions. Fluxes use shared faces and a conservative update:

$$V_{\text{next}}(\text{cell}) = V_{\text{now}} + \text{rain} + \text{sources} - \text{evaporation} - \text{infiltration} \\ + \text{sum}(\text{inflow_faces}) - \text{sum}(\text{outflow_faces}) - \text{withdrawals}$$

Compute each shared face flux once with equal and opposite contributions. Limit outflow by available volume, retain rounding residuals, and report every source/sink. A shallow-water or simplified diffusive-wave solver is a prototype choice with a stability test; a visual noise texture is not a flood model. Use adaptive substeps with a deterministic rule, or a fixed stable step at the worst permitted configuration. Any CFL-style timestep restriction is calculated from cell size and propagation speed in the same units. Refined/coarse boundaries conserve volume through accumulated face fluxes.

Pipes and surface drainage are different networks. Storm drains consume inlet capacity, pipe conveyance, pump power, and downstream outfall capacity. A blocked outfall can back up upstream. Drinking-water reservoirs, treatment, pumps, towers, and distribution account separately for volume and quality. Wastewater has collection and treatment capacity with bypass rules that create pollution. The initial hydraulic abstraction can use capacities and pressure zones; full transient fluid simulation is unnecessary unless its extra behavior improves decisions and survives the budget.

Electricity initially uses deterministic supply allocation on connected capacitated networks with line-loss approximations. This is a gameplay abstraction, not an AC power-flow simulator. Priority curtailment, storage charge/discharge limits, transformer bottlenecks, fuel, maintenance, and islanding create useful decisions. A later DC load-flow model requires explicit impedance data, numerical stability, and better UI explanations; it is not automatically a better game. Heating can similarly model generation, thermal storage, pipe capacity, losses, and temperature-dependent demand.

Air pollution uses source emissions, wind advection, diffusion, decay/deposition, and boundary exchange. Track pollutant mass where appropriate and integrate exposure over time; one frame of smoke does not injure an entire district. Noise is a separate intensity/exposure field. Sound levels cannot be added as ordinary linear decibel values; combine an explicitly linearized energy-like quantity before mapping to a displayed level. This is a simplified game field, with calibration and units documented rather than a claim of engineering-grade environmental assessment.

Weather is a seeded regional process with seasonal curves, bounded stochastic variation, and event timelines. Temperature affects heating, snow, road maintenance, and outdoor activity through named coefficients. Snow accumulation and clearance consume time and fleet capacity. Soil fertility, moisture, and crop calendars feed farms; forest growth, extraction, and fire fuel have renewable stock or depletion accounts. Mineral deposits are finite inventories with extraction grades and reclamation state. This adds regional specialization through stocks and costs rather than invisible production bonuses.

Disaster simulation composes normal failures: an earthquake damages edges and buildings; a flood blocks roads and contaminates supplies; a storm reduces airport and power capacity; a wildfire changes access and creates evacuation demand. Evacuation has destinations, shelter capacity, household reunification, vehicle availability, and return conditions. Difficulty options can disable hazards without removing the ordinary service and repair state machines.

9. Performance budgets that can fail visibly in development

All following values are proposed budgets for the reference workload and must be replaced or confirmed by measurement. A 20 Hz simulation has a 50 ms deadline at 1x. I would budget approximately 35 ms of critical-path wall time for a normal tick and preserve 15 ms for variance, OS scheduling, and burst work. An initial reservation is 12 ms transport/movement, 6 ms routing/network updates, 4 ms population/economy, 4 ms services/utilities, 3 ms environment, 3 ms commit/snapshot, and 3 ms diagnostics/other. These allocations sum to 35 ms; they are planning reservations, not measured independent p99 values that can be naively added. Capture the actual job graph critical path and total worker CPU time separately.

The presentation CPU has its own frame deadline, as does the GPU. CPU and GPU work overlap; adding all their elapsed times together is generally wrong. At the proposed 60 Hz visual tier, reserve a GPU p95 target below 13 ms and a presentation CPU critical path below 6 ms, while requiring actual p99 frame delivery below 16.67 ms in the certified scene set. Simulation and rendering still contend for memory bandwidth and cores, so the combined game must pass; isolated subsystem measurements are diagnostic only. At 3x, the simulation's available average time per tick is about 16.67 ms, so 3x is a workload-dependent capability rather than an unconditional guarantee at the largest city.

Proposed launch memory allocation on a 32 GiB machine is 16 GiB steady process commitment with a separate 4 GiB transient allowance, leaving room for the OS and ordinary background applications. Within the steady target: 3 GiB authoritative simulation and indices, 2 GiB routes/network caches, 5 GiB CPU asset/streaming data, 2 GiB engine/UI/audio, 2 GiB loaded mod reserve, and 2 GiB other/headroom. VRAM targets 9 GiB of a nominal 12 GiB budget with 3 GiB headroom for transient resources, drivers, and variance. Memory is not interchangeable across categories; monitor resident/committed CPU memory, GPU allocations, residency, and paging independently.

The arithmetic that informs storage choices is simple but valuable. One million 96-byte citizen core records consume 96 MB decimal before allocation/index overhead. Three transform snapshots for 100,000 visual entities at 32 bytes each consume 9.6 MB. One million 80-byte path requests consume 80 MB before frontier storage, which may be much larger. Keeping a 1 KiB history on every million residents consumes about 1 GB. Therefore keep detailed histories for selected/diagnostic entities plus bounded aggregated histories and sampled event traces, not unrestricted per-person logs.

Preallocate pools for common fixed-size records and arenas for bounded per-tick scratch data. Track allocator high-water marks, fragmentation, retained capacity, and burst paths. An 'allocation-free frame' is not sufficient if yesterday's route queue retained gigabytes. Pool exhaustion has a specified behavior: reject a new discretionary request with a pending reason, spill to a bounded secondary store, or stop advancing with an actionable resource warning. Never corrupt state or delete authoritative entities to reclaim memory.

9.1 Measure the right workloads

The performance corpus includes an empty map, a balanced 10,000-person town, the 250,000-person launch metropolis, the unadvertised million-person stress city, a dense downtown, dispersed suburbs, a transit-heavy city, a freight port, a deliberately broken road network, a large redevelopment, a disaster/evacuation, heavily modded assets, and a mature save with long history. Each fixture records active travellers, junction movements, building occupancy, freight volume, events/tick, graph changes, and cache hit rates.

Every run has a warm-up, a fixed command stream, a defined capture window, and repeat runs. Cold-cache loading and first shader use are separate tests. Report p50/p95/p99 tick/frame time, worst rolling window, achieved speed, memory high-water, allocation volume, route/service queue ages, GPU pass costs, and input-to-feedback latency. Compare energy/thermal throttling behavior on a sustained run. Disable unrelated benchmark noise where possible, then repeat under representative background load.

Optimize from captures: reduce needless updates; reduce data touched; improve locality; cache stable results with valid invalidation; partition independent work; simplify unnecessary fidelity; only then tune low-level instructions. A bottleneck can move after each change. Tracy supports CPU/GPU instrumentation and sampling, allocations,

locks, and context switches, making it a suitable candidate for the native kernel's diagnostic layer. Use the engine and platform GPU tools for the presentation side. [Tracy profiler](#).

10. Verification, debugging, and continuous integration

The test strategy has several independent questions. Is the state valid? Is the result reproducible? Does the simulation behave plausibly under its declared assumptions? Is the game understandable and enjoyable? Is it fast enough? A passing conservation test does not answer the other four.

Unit tests cover typed time conversions, integer rounding, safe arithmetic, handles, transactions, curve evaluation, predicates, graph edits, and controller transitions. Property tests generate random valid and adversarial networks and assert reachability, ownership, reservation exclusivity, volume/money reconciliation, and unchanged hashes after rejected commands. Differential tests compare optimized routing or utility allocation against a slow trusted implementation on small worlds. Metamorphic tests vary camera, render settings, worker count, entity allocation order, and irrelevant cosmetic assets while expecting identical authority.

Model tests use hand-computable fixtures. Two houses compete for one available clinic slot. A road with one blocked exit backs up. A supplier with ten units cannot fulfill two eight-unit orders. A building receives power through one saturated edge. A port with a full yard cannot unload indefinitely. A road demolition must release or legally remap all reservations. Test ordinary scenarios alongside adversarial ones so a repair does not make normal commuting worse.

Long-running tests exercise lifecycles and numerical accumulation: many simulated years, repeated save/load, repeated map editing, debt repayment, immigrant families, firm bankruptcy/reopening, and disaster recovery. A soak test checks both leaks and slow state growth: event tombstones, orphaned routes, missing subscription removal, ever-growing caches, and accumulated tiny money residuals. Run intentionally small caps to force recovery paths frequently.

Native memory safety uses compiler warnings, static analysis, fuzzing, and sanitizer configurations. AddressSanitizer detects classes of invalid memory access; ThreadSanitizer detects data races in supported builds. They are complementary and do not prove logic correctness. Run supported sanitizer jobs in separate configurations, with symbols and minimized reproductions attached. [Clang AddressSanitizer](#); [Clang ThreadSanitizer](#).

10.1 The eight-step debugging protocol

1. State the failure precisely. A bug record contains build and content hashes, OS/driver/hardware, ruleset, save/seed, logical tick, active mods, relevant entity IDs, command sequence, expected result, and observed result. A screenshot locates a symptom; a trace or reproducer establishes the state transition. Distinguish a recorded-video artifact from a live rendering defect, and simulation slowdown from low display FPS.

2. Preserve a reproducer. Capture the last checkpoint, command journal, mod lockfile, and a bounded pre-failure history. For nondeterministic presentation failures, capture the camera path, quality settings, GPU markers, and frame resources where possible. Reduce a city to the smallest causal subgraph without removing the failure; keep the original large save for performance context.

3. Draw the causal path. Map source event, owned state, queues, reservations, decision phase, transaction, snapshot aggregation, and UI formatting. For a silent landfill, inspect construction -> commissioning -> access -> staffing/utilities -> provider registration -> request assignment -> collection -> aggregation. The first red panel is not automatically the faulty owner.

4. Instrument without erasing the bug. Add counters, fixed-size history buffers, sampled spans, request/response IDs, state hashes, and queue-age histograms. Avoid logging every entity to disk each tick when investigating a timing bug. Measure instrumentation overhead and compare with a minimally instrumented build. A debug toggle should not accidentally change RNG streams or scheduling semantics.

5. Test competing explanations. For a factory with no deliveries, compare no supplier stock, no route, full terminal, missing access connector, expired reservation, slow route backlog, and stale UI. Each hypothesis gets a disproof experiment. A replay with unlimited route work that still fails weakens the backlog hypothesis. A panel

reading the same tick as the ledger can isolate a stale aggregation issue.

6. Repair the representation or invariant. If multiple service systems disagree about whether a building is active, introduce one lifecycle authority rather than separate booleans and UI patches. If shipments duplicate on reload, make custody transfer transactional and persist in-flight ownership. A local guard can prevent corruption immediately, but retain an explicit task to fix the ownership model if the guard only masks it.

7. Keep a regression. Add the minimized reproduction and the original workload to the appropriate suite. Check the exact failure, adjacent states, old saves, mods where relevant, pause/fast-forward, and worker variation. Tests should assert a meaningful outcome and invariant rather than copy the implementation's internal sequence.

8. Measure the cost. Capture CPU, memory, latency, queue ages, frame pacing, and changed gameplay outcomes before/after. A fix that prevents route invalidation errors by clearing every cache on every edit can be correct in a tiny fixture and unusable in a large city. Approval requires both correctness evidence and representative cost.

10.2 CI and release gates

Per change: build kernel/client adapters, unit/property fixtures, schema checks, content lint, a short deterministic replay, and a small performance smoke test. Nightly: long city replays, migration corpus, edit fuzzing, sanitizer jobs, cross-worker comparisons, render routes, and full benchmark scenarios. Milestone: external hardware matrix, accessibility sessions, creator/mod tests, cold-cache installs, offline/cloud conflict recovery, and long player sessions.

Performance gates compare against a pinned baseline with noise tolerances and repeat confirmation. A proposed 5% regression threshold in a stable benchmark should trigger investigation, not automatic concealment by changing the fixture. No release with known save-loss or conservation corruption, uncontrolled progression blockers, repeatable major graph corruption, or unresolved severe crashes. Visual defects are prioritized by prevalence and gameplay impact; a beautiful empty map does not waive a broken metropolitan benchmark.

Diagnostics are local by default. Optional crash/performance uploads minimize data, show what is included, and strip user paths and free-text names where feasible. Saves can contain personal city/building names and custom assets; obtain an explicit submission action before uploading a full reproducer. Online mod discovery and update services use their own reliability and security boundaries; a service outage must not prevent an installed offline city from loading.

11. Production organization, schedule, and cost model

The project is a multi-year team effort. I would plan around a 36-48 month initial release, subject to the research gates, rather than promise a date before the simulation/renderer spike. A smaller team must narrow the feature set and content volume. The critical path is integrated simulation correctness, tooling, and large-city performance; adding artists cannot shorten an unresolved road-topology or save-compatibility problem.

An illustrative peak team is 65 people: 18 engineers, 7 systems/game designers, 18 art/technical-art staff, 9 QA/automation specialists, 4 UI/UX/accessibility specialists, 3 audio specialists, and 6 production/build/community/content-operations staff. Some roles can be shared or external. Within engineering, explicitly staff simulation, transport/geometry, rendering/performance, tools/content, UI integration, persistence/platform, and build/test infrastructure. External localization, compatibility labs, legal/commercial work, and platform certification are additional scope.

A planning cost scenario uses an average of 50 full-time equivalents over 42 months: 175 person-years. At an assumed fully loaded annual cost of \$160,000-\$240,000 per person, labor is \$28-\$42 million. A further 25%-40% for external production, hardware/services, research overruns, and contingency produces roughly \$35-\$59 million before marketing and other excluded commercial costs. This is arithmetic from explicit assumptions, not a sourced estimate of Cities: Skylines II's budget or a market quote. Geography, compensation, outsourcing, engine terms, and scope can change it substantially; avoid double-counting overhead already in the loaded rate.

11.1 Milestones and exit evidence

Months 0-3: architecture and risk retirement. Build the headless tick kernel, stable IDs, typed units, command journal, profiler integration, save/load skeleton, and a minimal engine adapter. Demonstrate worker-independent hashes and camera independence. Spike dense rendering, road edits, geometric robustness, and route budgets. Select a provisional hardware matrix. Exit only with measured evidence for the engine and data model.

Months 4-9: the closed-loop town. One household type, employer, shop, dwelling, road network, utility, and service must produce a complete causal loop. Include money/goods ledgers, genuine trips, construction/activation, a useful inspection panel, save migrations, and a tiny mod. Build automated city generators and bad-network fixtures now. Exit with a playable town that survives a blockage, outage, bankruptcy, and reload without special-case rescue.

Months 10-18: vertical slice. A visually coherent 10,000-person town with public transit, industrial logistics, multiple zoning densities, terrain/water, seasons, municipal budgets, core services, audio, and accessible construction UI. The content pipeline creates final-quality assets and validates costs. Exit with end-to-end playtests, a known memory envelope, and no unresolved architecture-breaking defect.

Months 19-28: metropolitan scale and breadth. Grow the certified 250,000-person scenario; add rail/port/airport operations, broader policies, disasters, redevelopment, mature household/firm transitions, and regional specialization. Hybrid traffic only graduates after comparison against the baseline. The mod SDK and asset editor use the same contracts as internal content. Exit with performance gates on representative hardware and a fully populated migration/mod test corpus.

Months 29-36: alpha and feature lock. Finish all launch systems, scenario progression, tutorials, localization, accessibility, and content variety. Stop adding foundational features. Test years of city evolution, pathological networks, cold-cache installs, long sessions, missing mods, and cloud conflicts. Exit with stable saves, usable diagnostics, predictable recovery, and a performance margin.

Months 37-42, with reserve through 48: beta and release candidate. External playtests and hardware coverage drive repairs. Qualify platform builds, packaging, updates, rollback, support tools, and offline operation. Maintain a release branch while fixes continue on main. If the primary metropolitan gate fails, reduce scope, delay, or publish a clearly smaller supported envelope; do not redefine a broken city as a successful benchmark.

11.2 Concrete first 90 days

Weeks 1-2 establish repository boundaries, style/toolchain, reproducible builds, CI, logging schema, unit types, identity, and a benchmark ledger. Weeks 3-4 implement fixed ticks, deterministic random streams, event queues, command commit, hashing, and save/replay for a toy world. Weeks 5-6 create robust road edits, a simple lane graph, parcel frontage, and a single transport model. Weeks 7-8 close the household/shop/employer transaction loop and add inspectable utility/service states. Weeks 9-10 connect the renderer, camera, picking, road preview, and a constrained asset cooker. Weeks 11-12 profile synthetic and realistic workloads, inject failures, validate old saves, and make the engine/go/no-go decision with recordings and traces.

Deliverables include a runnable headless command such as `citysim run --save town --commands scenario --ticks N --hash-out result`, a deterministic city generator, a state-inspection tool, reference snapshots, a trace viewer configuration, geometry golden cases, and a content validation report. These are proposed interfaces, not claims that this report has implemented a game.

12. Governance and long-term maintenance

Maintain short architecture decision records for consequential choices: clocks, ownership, route hierarchy, simulation fidelity, engine version, numeric policy, save format, mod trust model, and supported scale. Each record names alternatives, evidence, reversibility, risks, and the next review trigger. A design change that affects a conserved quantity needs an invariant and migration plan before implementation.

Feature ownership spans design, engineering, art, UI, QA, and telemetry. A hospital feature is unfinished until it has usable placement, a commissioned lifecycle, access/staffing/utility requirements, actual patients, interpretable queues, save compatibility, audio/visual state, accessibility, and a performance case. This prevents 'simulation done' and 'UI done' from hiding an unusable joint product.

Post-launch work uses staged patches, opt-in previews, save backups, a compatibility matrix, and a published distinction between visual-only and simulation-changing updates. Diagnose from consented real saves as well as lab fixtures. Track crash-free sessions, save failures, queue starvation, late ticks, shader hitches, common confusion, and mod breakage. A new content pack enters the same performance and conservation gates as base content.

Potential multiplayer should be prototyped as a separate product constraint. A server-authoritative simulation could stream snapshots and accept commands without demanding every client simulate the whole city, but bandwidth, late joining, conflicting edits, ownership, pause/fast-forward, desync recovery, anti-abuse, and mod compatibility all require design. Deterministic replay helps debugging; it does not make multiplayer free. Do not distort the first single-player architecture around an uncommitted network feature.

13. The most consequential risks and decisions

- **Clock/scale mismatch:** long trips can consume an entire compressed day. Validate representative commute distributions early; adjust calendar scale, map accessibility, or activity cadence coherently.
- **Hybrid traffic divergence:** aggregation can erase lane behavior or queue order. Keep a simple reference model and explicit error bounds; reject optimization if it changes important decisions.
- **Economic instability:** reinforcing rent, migration, employment, and pricing loops can oscillate. Introduce explicit delays, bounded responses, scenario shocks, and inspectable contributions; do not hide defects with unlogged cash.
- **Geometry pathology:** arbitrary player edits create near-degenerate networks. Robust predicates, transactional compilation, saved seeds, and rejection that preserves old state are foundational.
- **Memory bandwidth:** more workers can make a sparse traversal slower. Measure locality and bytes touched before increasing concurrency.
- **Content inflation:** one expensive asset used thousands of times can dominate the city. Validate cost at expected repetition counts, not only in an isolated viewer.
- **Save/mod permanence:** public schemas become a long-lived contract. Keep authoritative definitions engine-neutral, versioned, and migratable from the first release.
- **Scope growth:** every mode and service multiplies interactions. Require an integrated slice before authorizing breadth; preserve the time and staff needed for regression and polish.

My release standard is a city that can become congested, insolvent, polluted, or underserved for understandable reasons; that can recover through sensible changes; that looks coherent from street to region; and that remains the same city after a camera move, a long session, a save, or a software update with a supported migration. That is the technical foundation I would build first and protect throughout production.

Appendix A. Implementation contracts and configuration

A.1 A feature is complete only across its boundaries

For each feature, produce a small contract with: owned state; permitted input commands; validation rules; input/output time domains; read dependencies; commit phase; authoritative events; resource transactions; UI queries; persistence schema; migration behavior; failure/recovery states; determinism guarantee; cost model; and regression fixtures. The same contract drives implementation review, content authoring, QA, and the inspector.

For example, a landfill owns its storage and processing capacity, while transport owns its vehicle movement and the garbage service owns collection requests. Pickup transfers waste from a building to a truck manifest; arrival transfers it to landfill storage; processing transfers it to a declared disposed/recycled sink. All three transfers have one receipt and one transaction owner. A panel computes collected-per-day from receipts, capacity-per-day from the operational model, and stored-tonnes from inventory. Those are different quantities and must never share an unlabeled number.

A.2 Proposed repository organization

/sim/core	identities, clocks, scheduler, transactions, hashes
/sim/transport	networks, routing, movement, transit, cargo
/sim/economy	accounts, households, firms, markets, development
/sim/services	requests, providers, dispatch, capacity allocation
/sim/environment	weather, water, emissions, damage
/sim/storage	schemas, checkpoints, replay, migrations
/client/adaptor	snapshots, selection, instance submission
/client/ui	semantic views, tools, overlays, localization
/tools	content cooker, map editor, validators, debuggers
/sdk	schemas, bindings, examples, mod packaging
/tests	tiny models, properties, replay and migration cases
/benchmarks	saves, commands, cameras, hardware manifests
/content	source definitions and source asset references
/docs	contracts, decisions, tuning and release procedures

The build graph forbids simulation modules from depending on client modules. Shared data schemas are generated into native code, editor views, and mod bindings. Generated files are reproducible and carry a schema hash. Schema evolution is reviewed as a product compatibility change, not merely a refactor.

A.3 Configuration and balance data

Define a versioned `WorldRules` object containing clock scales, economic coefficients, demographic transitions, routing work quotas, transport behavior profiles, service policies, environment solver parameters, difficulty assistance, and enabled features. Store it in saves by hash plus resolvable content version. Visual quality is a separate object that never changes authoritative rules.

Every tunable parameter records units, permitted range, default, rationale, expected monotonic effect, owner, and tests. Changing clinic capacity should usually increase potential treatment throughput; if a downstream shortage reverses that outcome, the inspector should show the bottleneck. Tune related parameter sets together. Avoid a thousand undocumented constants that individually look sensible but collectively create oscillations.

Sensitivity experiments vary one factor and then selected interactions: wage versus rent, transit fare versus parking price, utility cost versus industrial margin, vehicle capacity versus dwell time, and pollution versus moving cost. Record outcome distributions, recovery time, and player comprehension. A calibration target is allowed to be deliberately fictional when it makes planning clearer; mark it as such.

A.4 Operational state versus visual state

Visual objects carry their authoritative entity ID when applicable and an explicit decorative classification otherwise. Their disappearance due to culling is never a gameplay event. Render interpolation must handle birth, removal, demolition, and topology remap with discontinuity flags. A stale selected entity resolves to a tombstone explanation rather than a newly reused slot.

UI commands receive a unique command ID, expected world/network version, previewed cost range, and affected entity set. The simulation returns accepted/pending/rejected plus exact reasons. A pending road transaction cannot be committed twice by a double click; retries use the same idempotency key. Undo/redo and networked future clients use the same command semantics.

A.5 Bounded failure policies

Each queue declares maximum count, maximum item size, age policy, priority policy, and overload response. Critical simulation work does not get silently dropped. Cosmetic particles can be culled; a pending ambulance call cannot. A route queue may delay departure with visible waiting, but it cannot invent a route. A mod exceeding deterministic fuel cannot partially post payments. A corrupt asset can be quarantined; a missing authoritative definition cannot be quietly replaced by a random building.

Each cache declares its key, dependency epochs, memory ceiling, eviction order, rebuild cost, and whether rebuilding may affect behavior. Pure caches can disappear without changing results. Cached forecasts that

intentionally lag are model state, not disposable implementation detail; their observation window and last update belong in the save or deterministic reconstruction.

Appendix B. Terms and final decision checklist

Authoritative state: the data that determines what happens in the city. **Snapshot:** an immutable published view of a completed state. **Tick:** one fixed logical simulation step. **SoA:** structure-of-arrays, grouping like fields for efficient batch processing. **ECS:** entity-component-system organization; a data layout and processing pattern, not a performance guarantee. **LOD:** level of detail, which must distinguish visual detail from simulation representation.

Invariant: a condition that every legal transition preserves. **Conservation:** an account of where a tracked quantity originated, moved, transformed, or left the modeled system. **Reservation:** an exclusive or capacity-limited claim on a resource before final transfer. **Backpressure:** making new work wait when downstream capacity is full.

Spillback: a traffic queue physically extending upstream and blocking other movements. **Epoch:** a version identifying the inputs used by a derived structure or cache.

Critical path: the chain of dependent work that determines completion time. **p99:** the value below which 99% of the sampled measurements fall; the measurement window and workload still matter. **Replay determinism:** identical authoritative evolution from the same supported starting state and commands. **Migration:** a versioned transform that allows an older saved world to continue under a new supported schema or ruleset.

Before full production, approve or revise the engine adapter, map/clock scale, first transport representation, authoritative ID policy, numerical policy, resource ownership, save/replay contract, supported mod capabilities, primary performance scenario, and staffing. Before release, demand evidence that the full loop works: build -> populate -> employ -> travel -> produce -> deliver -> pay -> consume -> maintain -> age -> redevelop -> save -> reload -> continue.

The most valuable first deliverable is a small town where that loop is correct, observable, and fast. The metropolis is the same system under a larger and more adversarial workload, supported by better representations and tools rather than a separate collection of exceptions.

Appendix C. Linked research and documentation

The sources below support the nearby attributed statements and provide further technical reading. The original architecture, algorithms as adapted here, numerical budgets, and production estimates remain proposals. Documentation URLs can evolve; pin the chosen software versions during implementation. Sources accessed for this report on September 26, 2026.

- 1 [official Cities: Skylines II feature overview](#)
- 2 [Epic MassEntity overview](#)
- 3 [Unity archetype concepts](#)
- 4 [Factorio optimization case studies](#)
- 5 [CGAL exact computation guidance](#)
- 6 [CGAL kernel predicates and constructions](#)
- 7 [FHWA horizontal and vertical alignment summary](#)
- 8 [ASAM OpenDRIVE 1.8.1 specification](#)
- 9 [road-linkage rules](#)
- 10 [Geisberger et al., *Exact Routing in Large Road Networks Using Contraction Hierarchies*](#)

- 11 [Delling, Pajor, and Werneck, *Round-Based Public Transit Routing*](#)
- 12 [official GTFS overview](#)
- 13 [Daganzo, *The Cell Transmission Model: Network Traffic*](#)
- 14 [SUMO traffic simulation theory](#)
- 15 [SUMO mesoscopic model](#)
- 16 [SUMO safety documentation](#)
- 17 [Kesting, Treiber, and Helbing, *General Lane-Changing Model MOBIL*](#)
- 18 [TRB Highway Capacity Manual overview](#)
- 19 [Helbing and Molnar, *Social Force Model for Pedestrian Dynamics*](#)
- 20 [UIC RailSystemModel](#)
- 21 [Citizen Simulation & Lifepath](#)
- 22 [Economy & Production](#)
- 23 [Zones & Signature Buildings](#)
- 24 [City Services](#)
- 25 [Electricity & Water](#)
- 26 [Economy 2.0, part 1](#)
- 27 [part 2](#)
- 28 [A demographic microsimulation model with an integrated household alignment method](#)
- 29 [Mobile phone records to feed activity-based travel demand models: MATSim for studying a cordon toll policy in Barcelona](#)
- 30 [Traffic AI](#)
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